

Internet Appendix for
“How Resilient Is Mortgage Credit Supply? Evidence from the
COVID-19 Pandemic”

ANDREAS FUSTER, AUREL HIZMO, LAUREN LAMBIE-HANSON,
JAMES VICKERY, AND PAUL S. WILLEN*

ABSTRACT

The Internet Appendix provides additional analyses, tables, and figures supporting the main text. It is organized as follows:

- Section I. Additional Evidence on Rates and Markups in the Conforming Market
- Section II. Additional Evidence on Capacity Constraints
- Section III. Additional Evidence on Fintech Lending
- Section IV. Additional Evidence on Nonbank Financial Constraints
- Section V. Additional Evidence on Competition and Shopping
- Section VI. Additional Descriptive Statistics
- Section VII. Analysis of Credit Supply in Riskier Segments

*Citation format: Fuster, Andreas Aurel Hizmo, Lauren Lambie-Hanson, James Vickery, and Paul S. Willen, Internet Appendix for “How Resilient Is Mortgage Credit Supply? Evidence from the COVID-19 Pandemic,” *Journal of Finance* [DOI String]. Please note: Wiley is not responsible for the content or functionality of any additional information provided by the authors. Any queries (other than missing material) should be directed to the authors of the article.

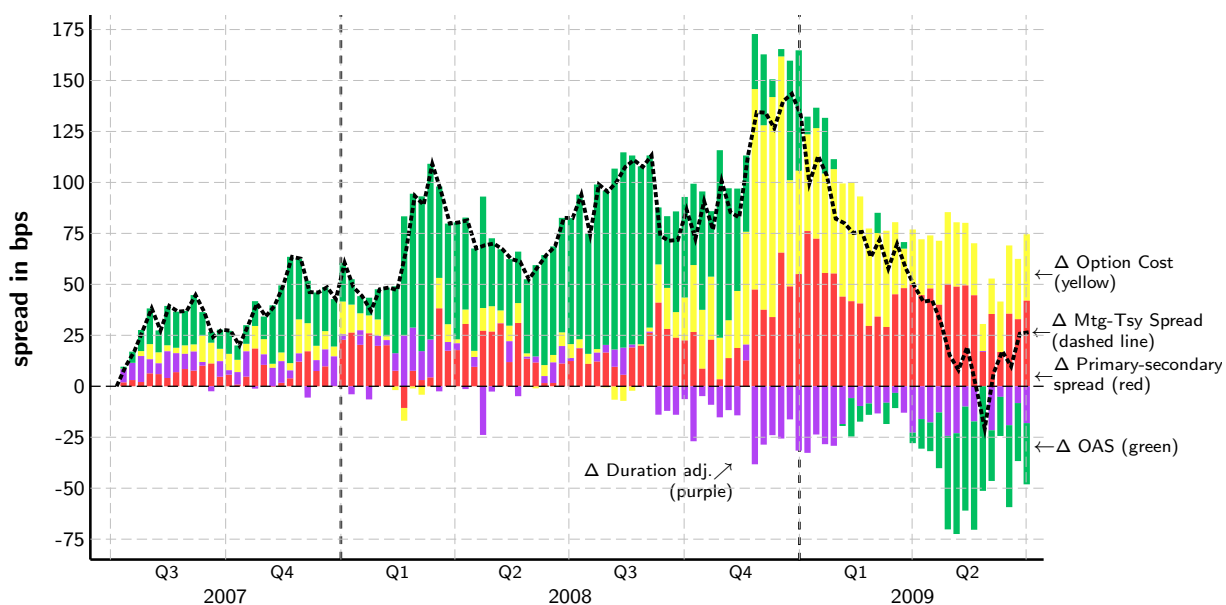
I. Additional Evidence on Rates and Markups in the Conforming Market

Figure IA.1. OAS measures from different dealers. OAS is calculated by interpolating values between the two MBS coupons on either side of the net note rate as described in Section II.A.1 of the main text. Data sources: Freddie Mac PMMS, J.P.Morgan Markets, Citi.

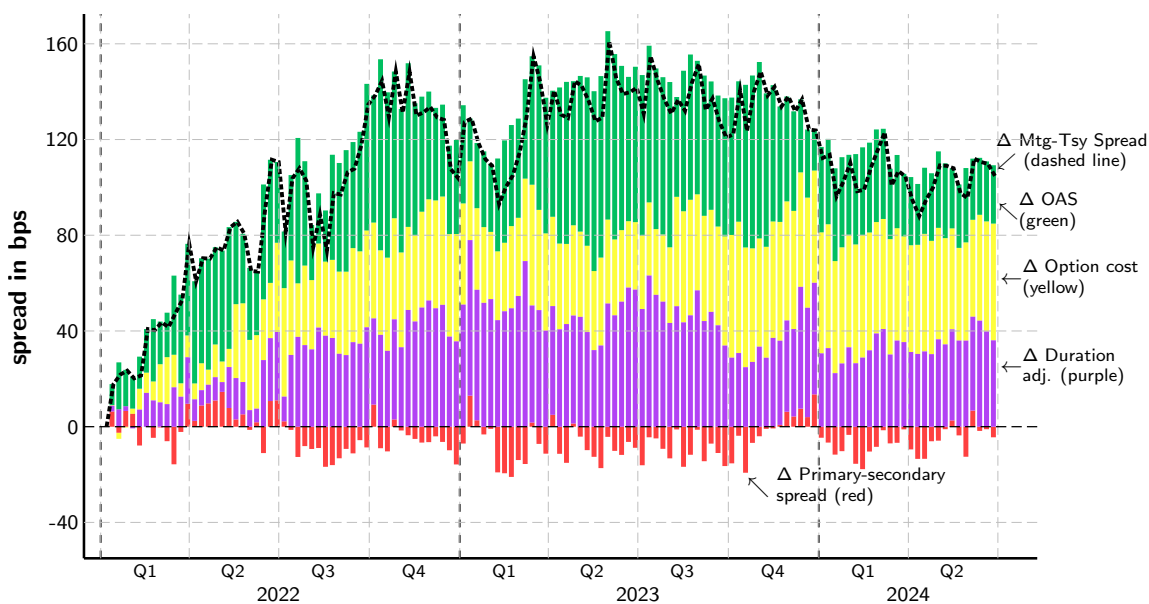


Figure IA.2. Tracing the mortgage-Treasury spread — other episodes. This figure decomposes the mortgage-Treasury spread into four components, based on the methodology described in Section II.A of the main text. The black dashed line shows the change in spread between the headline mortgage rate (from PMMS) and 10-year Treasury yields relative to the beginning of each episode, in basis points. Data sources: Freddie Mac PMMS, Optimal Blue, J.P. Morgan Markets.

A. Global Financial Crisis (2007-2009)



B. Monetary Tightening Cycle (2022-2024)



A. Details on the Gain-on-Sale Calculation

A.1. Formal Treatment

Consider a mortgage with initial balance S_0 and a mortgage rate r_p . Let r_s be the yield on an MBS containing mortgages with note rate r_p , and g be the guarantee fee paid to the GSEs. If we assume that time is continuous and the loan has a constant prepayment hazard λ_p , the value of the loan is:

$$V(r_p - g) = \int_0^\infty e^{-r_s t} S_t (r_p - g + \lambda_p) dt = S_0 \frac{r_p - g + \lambda_p}{r_s + \lambda_p}. \quad (\text{IA.1})$$

where the second equality follows from the fact that with a constant prepayment hazard, $S_t = S_0 e^{-(\lambda_p)t}$. Equation (IA.1) implies that the value of the mortgage, all else equal, is increasing in the note rate and decreasing in the g-fee and the MBS yield, as one would expect. Prepayment speed, λ_p , appears in both the numerator and the denominator and its effect is therefore ambiguous. Taking the derivative of equation (IA.1) with respect to λ_p yields

$$\frac{\partial V}{\partial \lambda_p} = -\frac{(r_p - g - r_s)}{(r_s + \lambda_p)^2} S_0.$$

Intuitively, if $r_p - g - r_s$ is positive, investors are receiving cash flow in excess of their funding costs (or outside investment opportunity, which is r_s if they invest in MBS) and would like to delay prepayment as long as possible. In general, $r_p - g - r_s$ is positive at origination and thus newly originated loans with lower prepayment speeds fetch a higher valuation in the market. For seasoned loans, $r_p - g - r_s$ can be negative (if market yields increased since origination), in which case higher prepayment speeds lead to higher valuations.

To compute gain-on-sale, we assume the lender can sell the loan in the MBS market for V . Then, using equation (IA.1), gain-on-sale as a share of the loan amount can be

expressed as

$$\frac{V(r_p - g) - S_0}{S_0} = \frac{r_p - r_s - g}{r_s + \lambda_p}. \quad (\text{IA.2})$$

Equation (IA.2) omits two components that are important in actual gain-on-sale calculations. First, lenders typically value the servicing flow s separately from the rest of the cash flow. The net servicing income flow is $s(1 - c_s)$, where c_s is the cost of servicing as a share of servicing income. Second, the lender typically collects points and fees at origination, which we denote as F , so the initial cash outlay is not S_0 but $S_0 - F$.¹ Taking those two factors into account, we can write gain-on-sale as

$$\underbrace{\pi}_{\text{Gain-on-Sale}} = \underbrace{\frac{V(r_p - g - s) - S_0}{S_0}}_{\substack{(1) \text{ Secondary Market} \\ \text{Income}}} + \underbrace{s \cdot \frac{1 - c_s}{r_s + \lambda_p}}_{\substack{(2) \text{ Mortgage} \\ \text{Servicing Right} \\ \text{(MSR)}}} + \underbrace{\frac{F}{S_0}}_{\substack{(3) \text{ Points} \\ \text{and Fees}}}. \quad (\text{IA.3})$$

$\frac{1 - c_s}{r_s + \lambda_p}$ is known as the servicing multiple.

A.2. Empirical Implementation

Our approach to implementing equation (IA.3) follows the principles established in Fuster, Lo, and Willen (2024).

For our baseline estimates, we assume r_p equals the Freddie Mac PMMS 30-year conforming rate for the time period in question. In Figure IA.3, we compare those estimates with estimates using the rates from the MBA weekly application survey and the Optimal Blue Insight data. Specific details of our calculations are described below:

1. For Secondary Market Income (component (1)), we compute $V(r_p - g - s)$ as follows. s and g are 25bp and 42bp, respectively, following Fuster et al. (2013). “TBA” stands for “to be announced,” representing the MBS market where new

¹ F is net of loan-level price adjustments (LLPAs) paid to the GSE when selling the loan.

agency mortgages are typically forward-sold (see Vickery and Wright, 2013). Given that MBS coupons are at 50bp intervals, interpolation between the two nearest MBS coupons is necessary to price a specific loan. For instance, if $r_p - g - s$ is 3.25%, the equal-weighted average price of a 3.0 coupon and a 3.5 coupon is taken. Further, as in Fuster et al. (2024), we assume that settlement happens 45 days later and use a weighted combination of prices on one-, two-, and three-month-out TBA contracts.

2. To compute the value of the MSR (component (2)), our challenge is to find empirical values for $\frac{1-c_s}{r_s+\lambda_p}$, the servicing multiple. To do this, we use data from SitusAMC, an independent valuation service company. These multiples are derived from transaction values of brokered bulk MSR deals, market participant surveys, and a pricing model, encapsulating the net value of servicing accounting for the income flow value minus expected costs.² The multiples are provided to us at monthly frequency at the coupon-by-loan type (GSE versus FHA) level. Figure IA.4 shows the evolution of these multiples for conventional conforming and FHA segments. Multiples for given fixed coupons decreased over time, reflecting the drop in market interest rates. However, the interpolated multiple at the typical coupon rate for new loans decreased only modestly before fully reverting over the course of 2020.³

²Fuster et al. (2013) calculate a measure of gain-on-sale called “OPUC,” where they explicitly consider the choice of coupon into which a loan is securitized but for the most part assume constant multiples over time. In Appendix I.B, we show that results using OPUC are qualitatively similar.

³Our estimates could overstate gain-on-sale if the SitusAMC servicing multiples fail to properly reflect the risks for servicers of elevated forbearance and non-payment. But in Section IV.B we show there is no evidence of a rising interest rate premium for high-risk conventional conforming loans during the pandemic, which speaks against this hypothesis. We also have no reason to doubt that the SitusAMC valuations internalize the risks of forbearance; in fact, given secondary market illiquidity

3. Finally, for each of our rate series, we construct a corresponding time series for Points and Fees (component (3)). Freddie Mac, in addition to collecting rate information, also gathered Points and Fees data from lenders—although they discontinued this practice at the end of 2022, which is after our period of interest. The Mortgage Bankers Association application data also include points and fees. For the Optimal Blue Insight data, we use their cross-sectional rate-point combinations to construct a time series with a constant level of points. We subtract a Loan-Level Price Adjustment (LLPA) paid to Fannie Mae and Freddie Mac for our baseline mortgage of 50bp and we further include the 25bp surcharge that FHFA imposed on refinances delivered to Fannie Mae and Freddie Mac after December 1, 2020.

Although the calculations are similar, for their baseline estimates, Fuster, Lo, and Willen (2024) use a different r_p series created using proprietary TPO rate sheet data, different servicing multiples, and by subtracting a 100bp rebate (which can be thought of as their calculation not including origination charges paid by borrowers). As a result, for the overlapping period, their estimates in levels are different but the intertemporal patterns are very similar.

Finally, our treatment of FHA loans is identical to what is described above, except that s and g are 44bp and 6bp, respectively, and no LLPAs are imposed.

for MSRs, transactions may understate going-concern values if they mainly reflect forced sales at fire-sale prices. Finally, even if we set the servicing multiple to zero—a very extreme assumption—this would reduce gain-on-sale by only about \$1, still not bringing it fully in line with historical patterns.

Figure IA.3. Intermediation markups in the conforming market. Primary-secondary spread (Panel A) and gain-on-sale (Panel B) measured based on the methodologies described in Sections II.A.1 and II.C of the main text, respectively. Data sources: Freddie Mac PMMS, Optimal Blue, J.P. Morgan Markets, MBA (via Haver Analytics). Vertical line represents the declaration of a national state of emergency on March 13th, 2020.

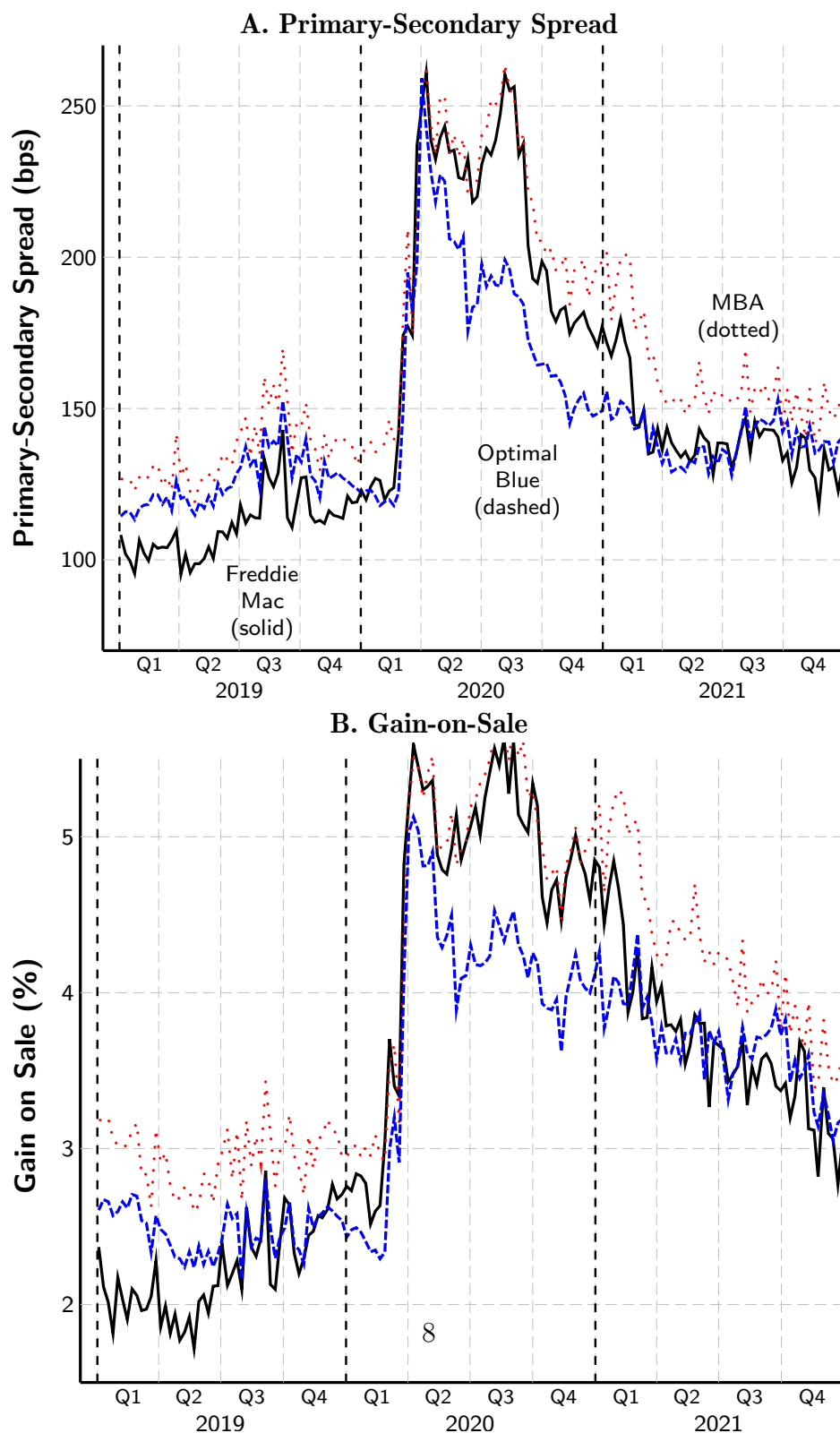
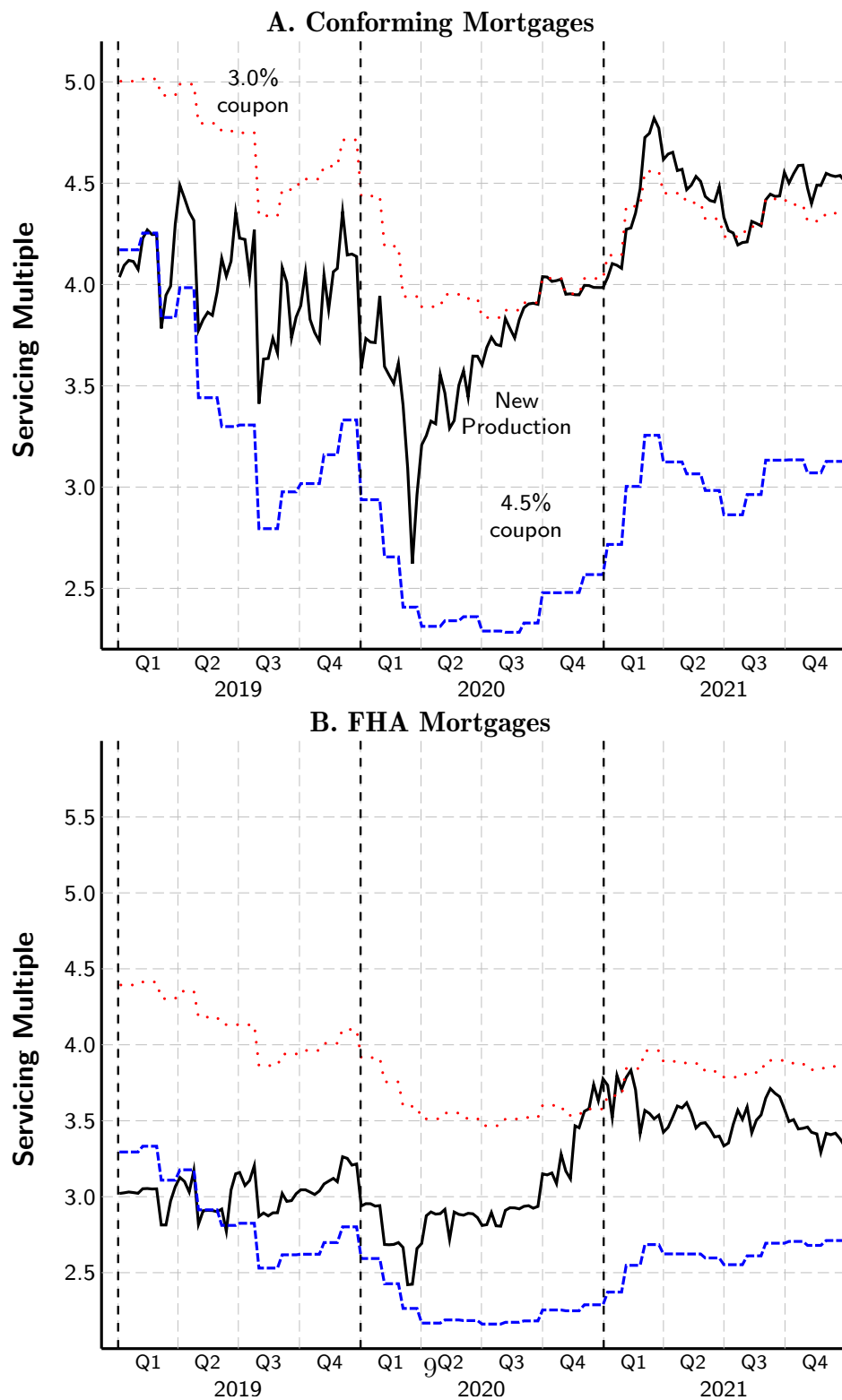


Figure IA.4. Base servicing multiples for conforming and FHA mortgages. “New Production” interpolates between specific coupons (two examples of which are shown) to obtain servicing multiples for a loan originated at the primary market mortgage rate adjusted for discount points. Data sources: SitusAMC for servicing multiples; Freddie Mac PMMS (Panel A) and MBA (Panel B) for mortgage rates.

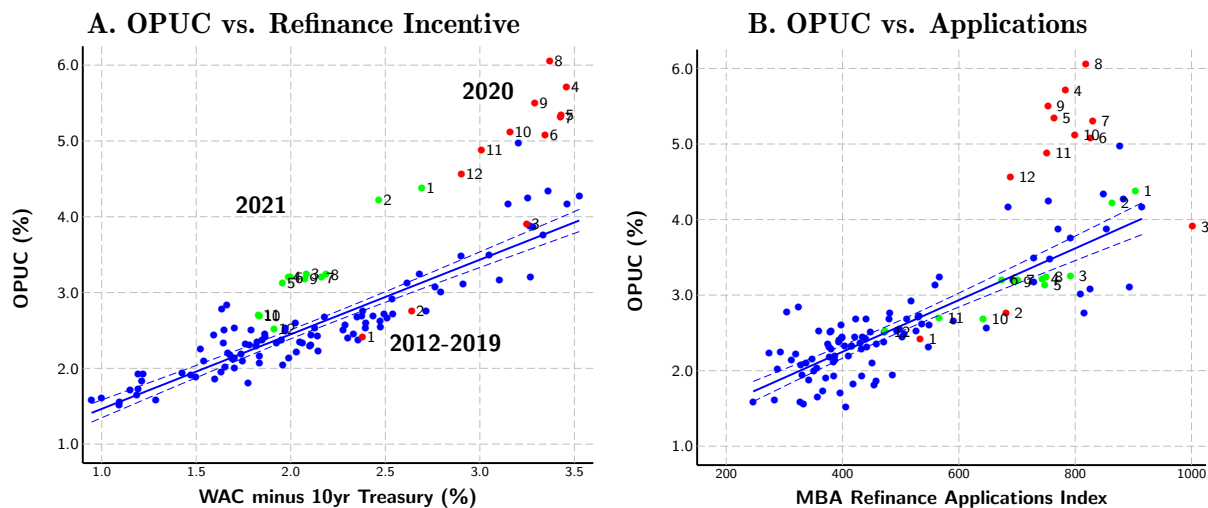


B. OPUC Measure of Intermediation Markups

Figure IA.5. Evolution of originator profits and unmeasured costs (OPUC). Data source: <https://www.newyorkfed.org/research/epr/2013/1113fust.html>. Vertical line represents the declaration of a national state of emergency on March 13, 2020.



Figure IA.6. OPUC and mortgage demand. Data sources: Freddie Mac PMMS; for OPUC: <https://www.newyorkfed.org/research/epr/2013/1113fust.html>; Mortgage Bankers Association (via Haver Analytics).



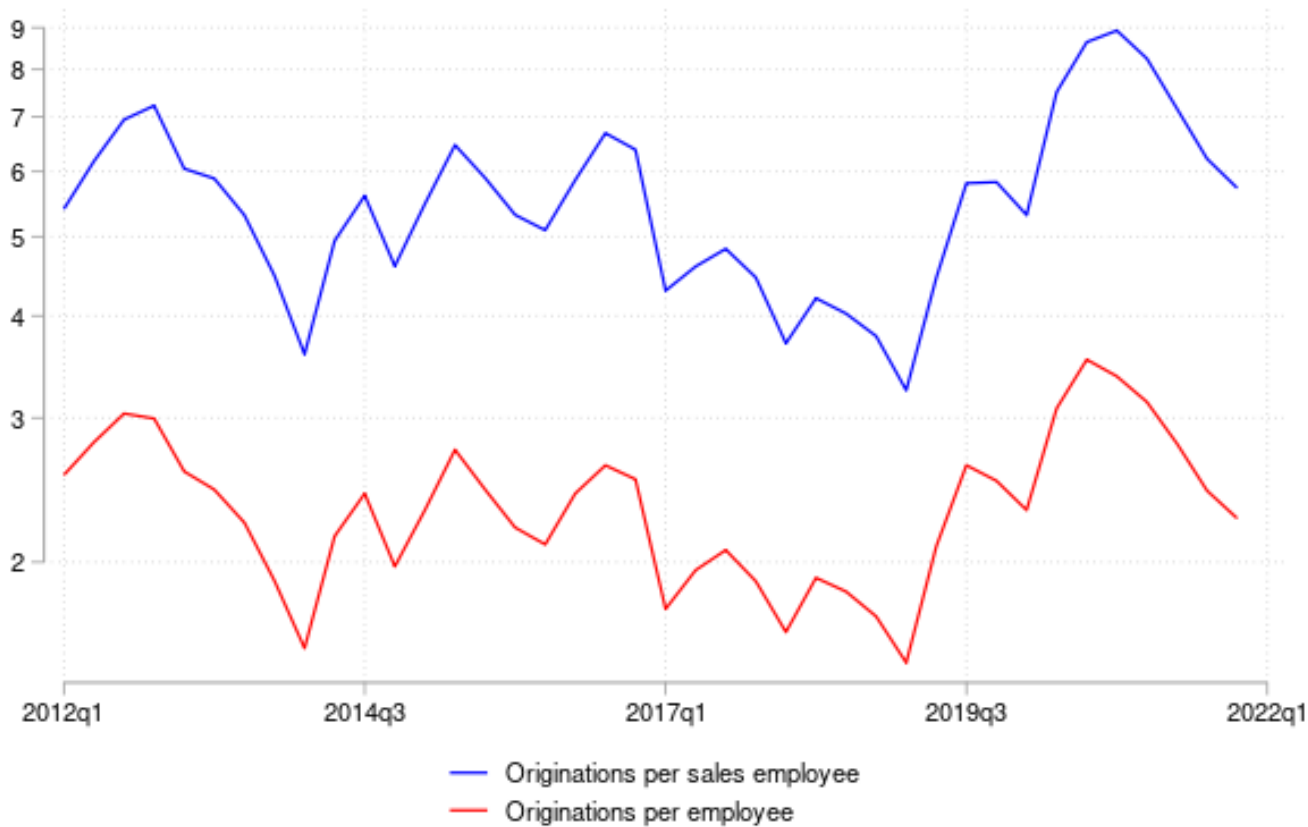
II. Additional Evidence on Capacity Constraints

Table IA.I. Time-Series Model of License Issuance. Monthly time-series regression of log(licenses) on lags of log(applications) and a seasonal dummy for December. License issuance is drawn from the NMLS; mortgage applications are from the Mortgage Bankers Association. Newey-West standard errors with 3 lags in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Excluding Pandemic Period		Including Pandemic Period	
	(1)	(2)	(3)	(4)
L.ln(applications)	-0.122 (0.080)	-0.090 (0.079)	-0.159 (0.130)	-0.109 (0.118)
L2.ln(applications)	0.259** (0.113)	0.242** (0.107)	0.282** (0.126)	0.169 (0.149)
L3.ln(applications)		0.051 (0.112)		-0.070 (0.139)
L4.ln(applications)		-0.045 (0.096)		0.263 (0.183)
December	-0.292*** (0.032)	-0.289*** (0.033)	-0.231*** (0.062)	-0.275*** (0.057)
Pandemic			-0.218* (0.125)	-0.281** (0.125)
Num obs.	59	57	69	67
Mean of dep. var.	8.64	8.65	8.62	8.62
SD of dep. var.	0.14	0.13	0.17	0.17
Adj. R2	0.33	0.39	0.21	0.29

Figure IA.7. Originations per employee – further evidence. Panel A shows the mean monthly originations per sales employee and per (total) employee using Mortgage Bankers Association Quarterly Performance Report data. Panel B shows the personnel costs for non-sales employees as a share of dollars originated, captured each quarter from 2012 to 2021.

A. Originations per Employee (Sales and Non-Sales Employees)



B. Personnel Costs (Non-Sales Employees)

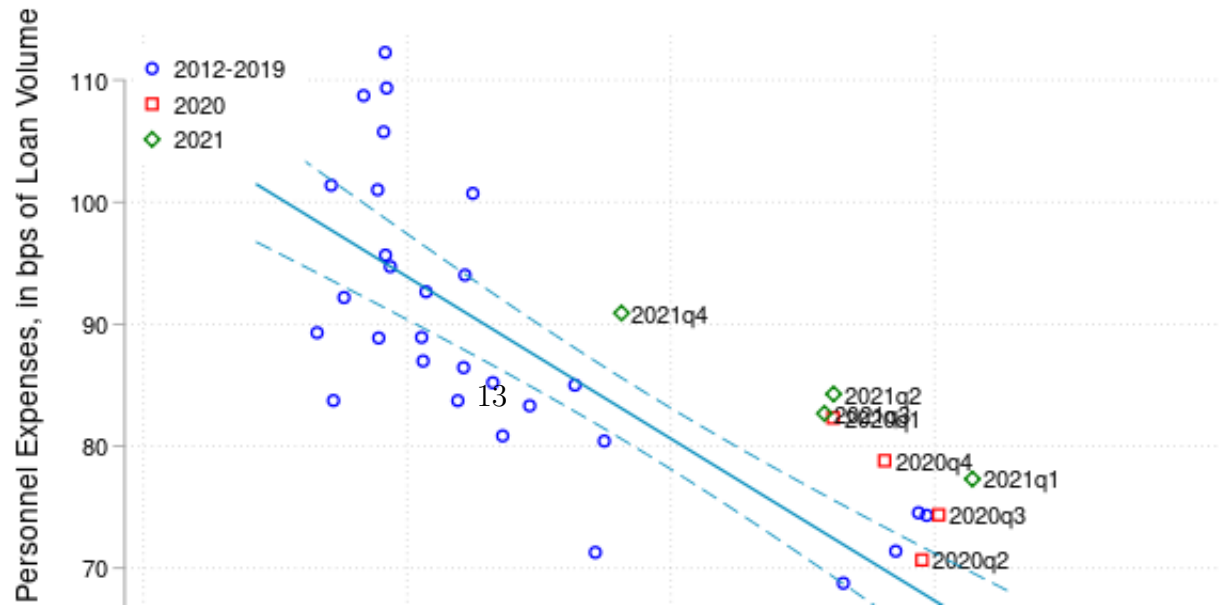


Figure IA.8. Median processing times and application volumes: 2012-2021. Data source: confidential-use HMDA data and Mortgage Bankers Association (MBA Application Index). Time is indexed by application date. Processing time is calculated for originated loans only.

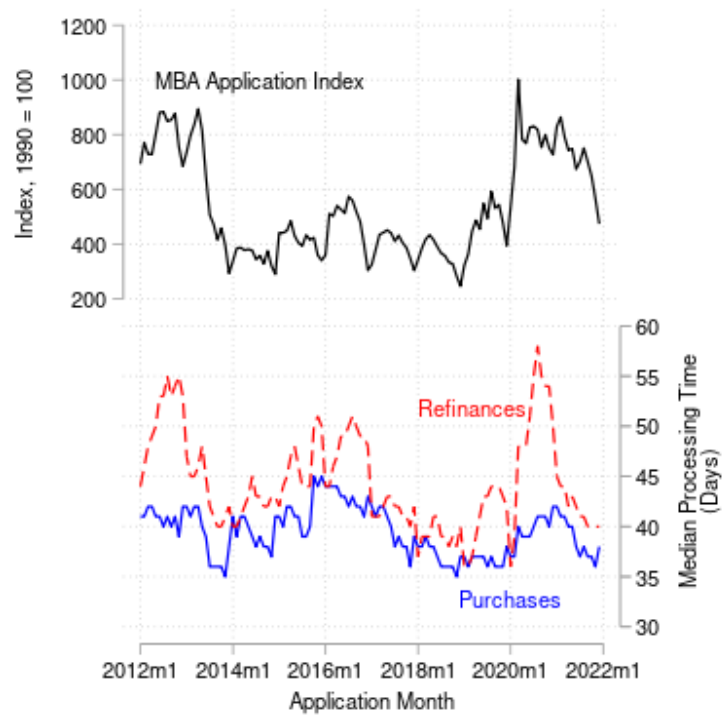
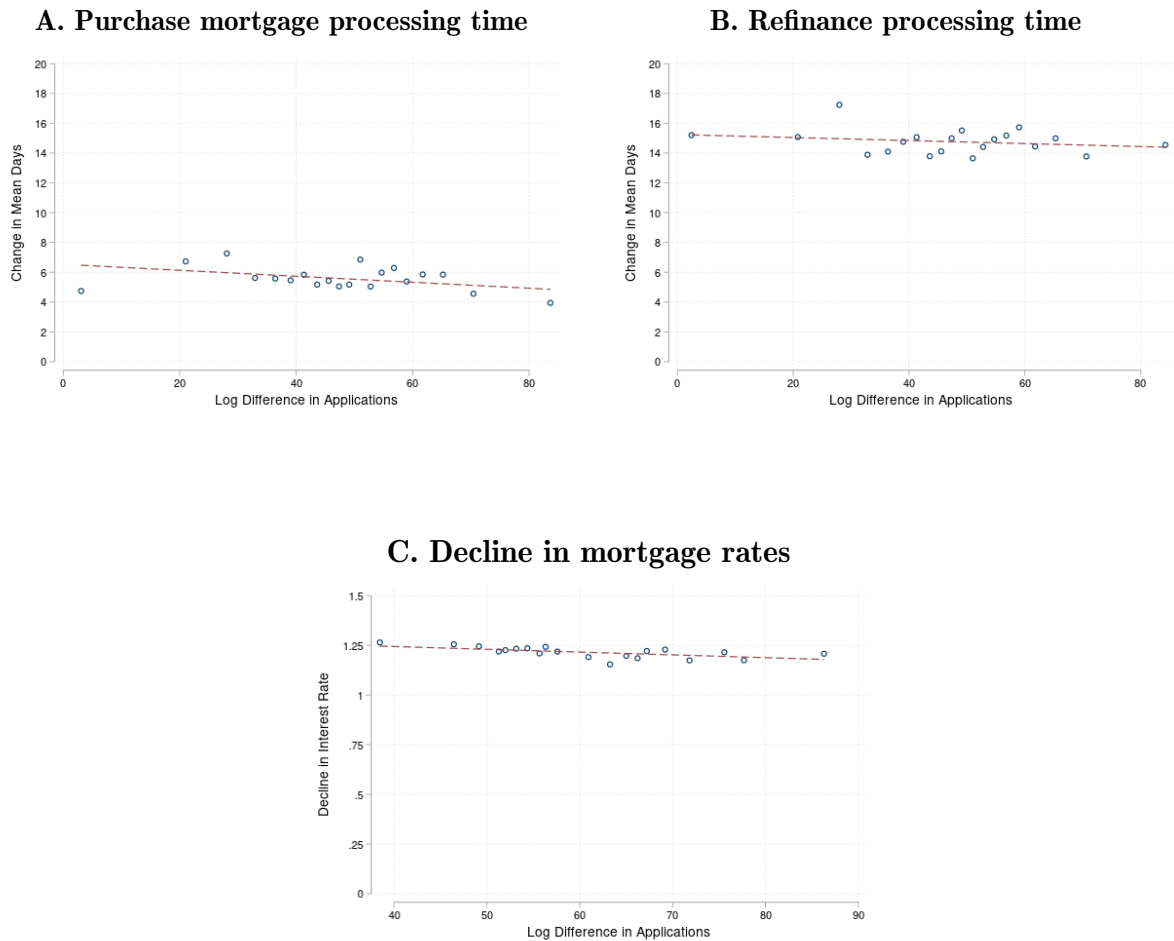


Figure IA.9. CBSA-level application growth, loan processing time increases, and rate declines. Binned scatterplot of CBSA-by-month, year-over-year log difference in mortgage applications from confidential-use HMDA data, displayed against mean processing time (in winsorized days) in Panels A and B and against mean decline in zero-discount-point interest rates from Optimal Blue Insight data in Panel C. All panels display data for March-December 2020, controlling for calendar month.



III. Additional Evidence on Fintech Lending

A. *Fintech Processing Speed Over Time*

This section contains evidence supporting Section III.D of the main text, examining how the processing-speed advantage of fintech lenders relative to other nonbanks has evolved over time.

We run the following linear model using loan-level confidential-use HMDA data:

$$\text{ProcessingTime}_i = \beta_t \text{Fintech}_i + \gamma_{ct} + \Gamma X_i + \varepsilon_i \quad (\text{IA.4})$$

where ProcessingTime_i is the difference between the action date and application date (in calendar days) for originated loans recorded in HMDA, and the estimates of interest are the elements of β_t , a vector of coefficients on month dummies interacted with a dummy for whether the originator is a fintech lender based on the classification in Jagtiani et al. (2021). Controls include month $\times$ CBSA dummies (γ_{ct}) and X_i , which is a vector of loan and borrower characteristics that mimics the set of controls in Table 3 of Fuster et al. (2019); we control for $\log(\text{loan amount})$, $\log(\text{applicant income})$, a pre-approval dummy, and indicators for occupancy status, property type, applicant gender and coapplicant status, as well as dummy variables for missing values of the above variables. Time is indexed by application month. Standard errors are double clustered by lender and month.

We restrict the sample to conventional conforming mortgages, defined as mortgages below the relevant local conforming limit, which are not flagged as government mortgages (e.g., FHA or VA loans), because our focus is the conforming market and because mortgage processing and underwriting procedures vary across conforming, government, and jumbo loans. We further restrict the sample to originations by nonbanks; this creates a more “apples-to-apples” comparison because banks and nonbanks differ on other dimensions aside from technology, including regulation, funding, and

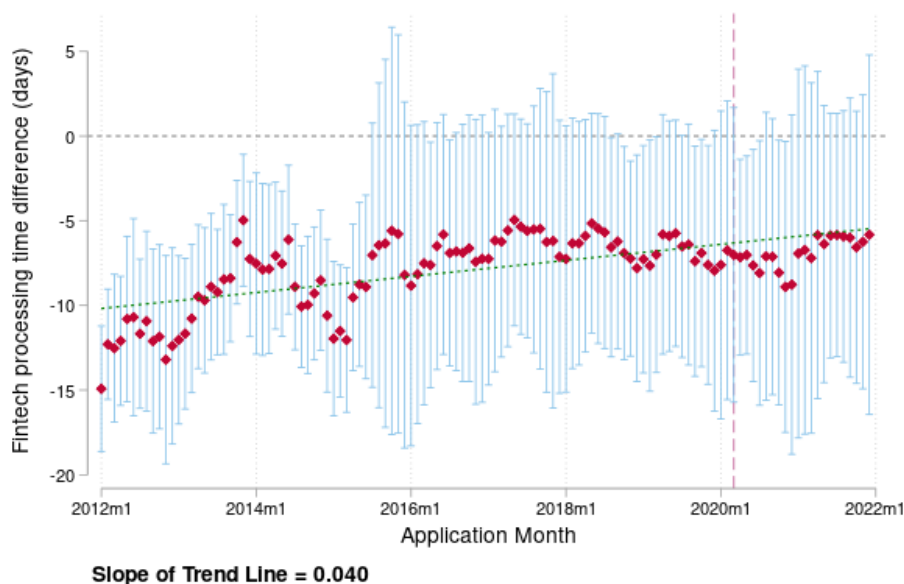
diversification across business lines. We estimate separate models for purchase mortgages and refinances given that the relationship between processing time and borrower characteristics may be different between the two types of loans.

Dynamic estimates of conditional differences in processing times between fintechs and other nonbanks are reported in [Figure IA.10](#). Throughout the sample period, fintech lenders typically process mortgages faster, although the advantage of these firms has diminished over time. Specifically, based on the trendlines, fintech processing speeds were about 10-12 days faster in 2012-13, similar to estimates in Fuster et al. (2019). Over the period from 2012-21, however, this “fintech advantage” declined by roughly 0.5 days per year (0.04 days per calendar month, which is the trend line slope reported in the figure) for purchase mortgages and 0.55 days per year (0.046 days per month) for refinances.

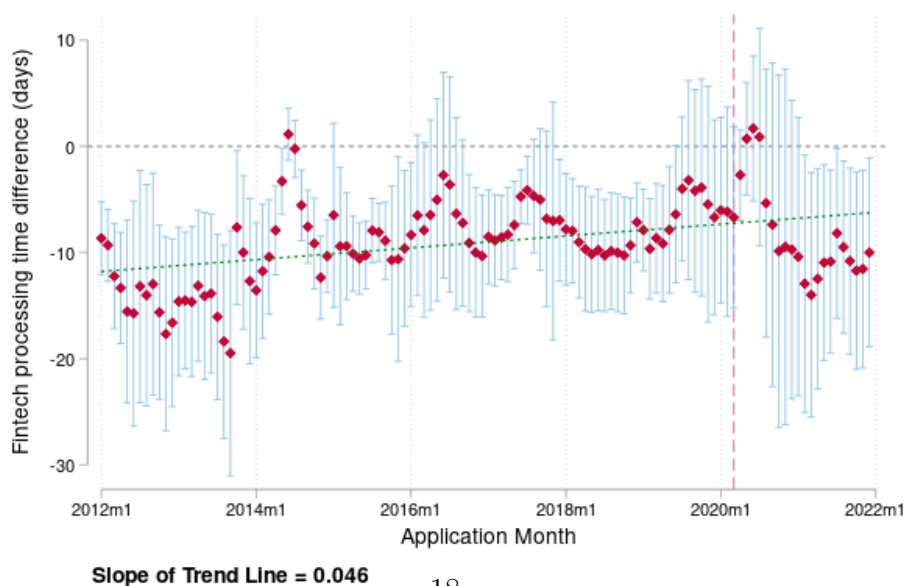
Fintech processing was however still about 6-7 days faster than other nonbanks as of late 2019 and early 2020 just before the pandemic, consistent with the argument that fintechs retain a technological advantage over other nonbanks despite the broader diffusion of online lending.

Figure IA.10. The evolution of the processing time advantage of fintech lenders. Figure shows the difference in processing time (in days) between fintech lenders and other nonbanks conditional on loan and borrower controls. Each point on the graph reports the coefficient on the fintech dummy (along with corresponding 95% confidence levels) from a loan-level linear model of processing time estimated separately month-by-month from January 2012 to December 2021 using confidential-use HMDA data and conditioning on the controls from Table 3 of Fuster et al. (2019). We estimate results separately for purchase mortgages and refinances, and restrict the sample to conventional conforming first-lien mortgages. Fintech lenders classified by the Jagtiani et al. (2021) classification. Dashed vertical line is March 2020.

A. Purchase mortgages



B. Refinances



B. Fintech Market Shares and Processing Times by Loan Purpose

Figure IA.11. Evolution of fintech market share of nonbank lending. Monthly coefficients and 95% confidence intervals showing the fintech share of nonbank conventional conforming mortgage lending. Based on regressions of a fintech originator dummy on time dummies and loan and geographic controls like the specification in column (3) of [Table III](#) in the main text, but with monthly dummies instead of a single pandemic dummy (using January 2020 as the base category). Time is indexed by mortgage application date. Standard errors clustered by county.



Table IA.II. Fintech Lending by Loan Purpose. This table repeats the analysis from Table III in the main text after restricting the sample first to purchase mortgages and then to refinances. Linear probability models estimating conditional changes in the fintech share of conforming mortgage lending during the pandemic as well as changes in the difference in mortgage processing time between fintechs and other mortgage lenders. Time is indexed by application date. Sample period is July 2019 to December 2020. Pandemic is defined as the period from March 2020 onwards. Loan controls include dummies for cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI², credit score, credit score², LTV, LTV², bins of applicant age, and dummies for missing values of each variable. Conforming loans are identified as mortgages that: i) do not exceed the relevant conforming loan limit and ii) are not flagged as government loans. Data source: confidential-use HMDA data. Standard errors clustered by county. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

A. Purchase Mortgages							
	=100 if lender is fintech; 0 otherwise				Processing time (days)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Pandemic	0.58*** (0.07)	-0.23* (0.09)	-0.22* (0.10)	-0.34*** (0.10)	4.09*** (0.13)	4.02*** (0.18)	3.52*** (0.18)
Pandemic x FICO<680				1.55*** (0.20)			
Fintech					-6.80*** (0.37)	-6.05*** (0.48)	-6.11*** (0.47)
Pandemic x Fintech					-0.32 (0.30)	-0.25 (0.33)	-0.30 (0.33)
Num obs.	4362054	2446561	2446183	2446183	4362054	2446561	2446183
Mean of dep. var.	9.26	16.51	16.51	16.51	50.08	48.98	48.98
Lenders	All	Nonbank	Nonbank	Nonbank	All	Nonbank	Nonbank
Loan controls	N	N	Y	Y	N	N	Y
B. Refinances							
	=100 if lender is fintech; 0 otherwise				Processing time (days)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Pandemic	1.81*** (0.18)	-0.74*** (0.21)	-0.15 (0.21)	-0.37 (0.22)	12.66*** (0.23)	12.56*** (0.18)	13.73*** (0.17)
Pandemic x FICO<680				3.27*** (0.29)			
Fintech					-10.65*** (0.29)	-4.33*** (0.21)	-5.23*** (0.20)
Pandemic x Fintech					0.02 (0.29)	0.11 (0.25)	0.24 (0.25)
Num obs.	8847778	5315568	5315223	5315223	8847778	5315568	5315223
Mean of dep. var.	19.42	32.32	32.32	32.32	56.90	51.49	51.48
Lenders	All	Nonbank	Nonbank	Nonbank	All	Nonbank	Nonbank
Loan controls	N	N	Y	Y	N	N	Y

IV. Additional Evidence on Nonbank Financial Constraints

Figure IA.12. Share of third-party originations. Share of third-party originations in the conforming mortgage market plotted against application date, constructed using confidential-use HMDA data. Conforming loans are identified as mortgages not larger than the relevant conforming loan limit that are not flagged as government loans. Third-party originations are measured based on a HMDA indicator variable for whether the borrower on a given loan submitted their application through a third party rather than directly to the lender. Vertical line indicates the onset of the pandemic, defined as the declaration of a national state of emergency on March 13, 2020.

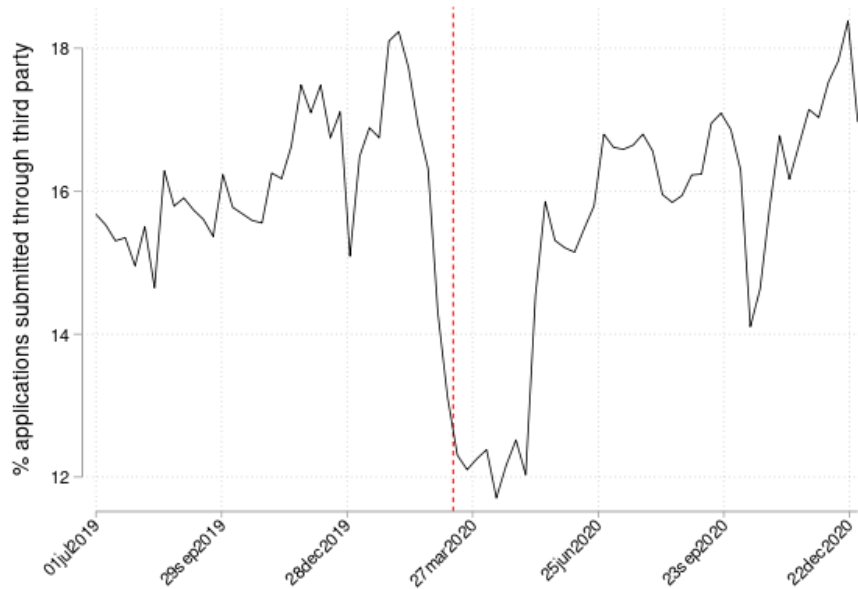


Table IA.III. Nonbank Share of Lending: Effect of Controlling for Third-Party Channel. Loan-level linear probability models showing conditional changes in the nonbank share over different phases of 2020. Column 1 reproduces column 3 from [Table IV](#) in the main text, while column 2 uses the same specification but includes a third-party-origination dummy as an additional control. Time is indexed by application date. Sample period is December 2019 to December 2020. Data source: confidential-use HMDA data. Standard errors clustered by county. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
Early pandemic [Feb 15-Mar 12]	1.31*** (0.15)	1.80*** (0.16)
Nonbank stress [Mar 13-Apr 30]	0.11 (0.15)	0.90*** (0.15)
Stress easing [May-Dec]	6.85*** (0.21)	6.93*** (0.20)
Num obs.	10602575	10602575
Mean of dep. var.	60	60
Loan controls	Y	Y
County dummies	Y	Y
Channel dummy	N	Y

Figure IA.13. Nonbank market share: jumbo loans. Nonbank share of jumbo mortgage lending plotted against application date, constructed using confidential-use HMDA data. “No controls” plots the raw nonbank share of lending. “Only county FEs” plots the nonbank market share controlling for geography, estimated by regressing a nonbank dummy on time dummies and county dummies, then plotting the estimated time dummies. Similarly, “All controls” plots the nonbank share conditional on a larger set of controls, including dummies for refinancing and cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI^2 , credit score, $credit\ score^2$, LTV, LTV^2 , bins of applicant age, county dummies, and dummies for missing values of each variable. Jumbo loans are identified as mortgages that i) exceed the relevant conforming loan limit and ii) are not flagged as government loans. Vertical line indicates the onset of the pandemic, defined as the declaration of a national state of emergency on March 13, 2020.

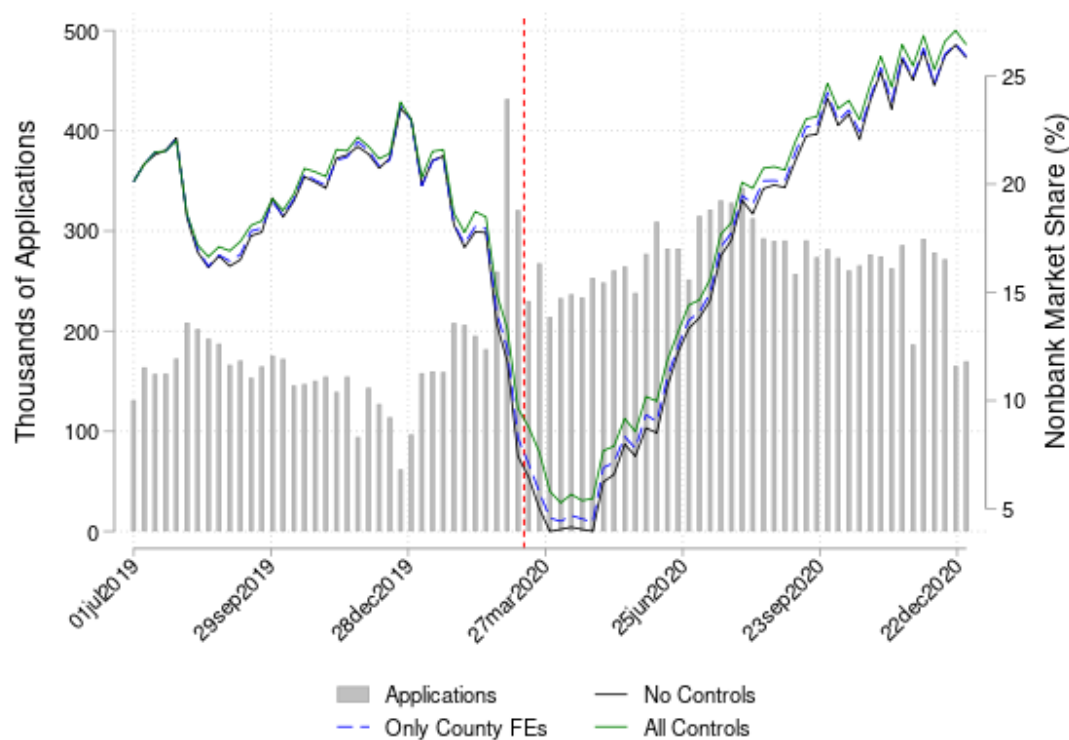
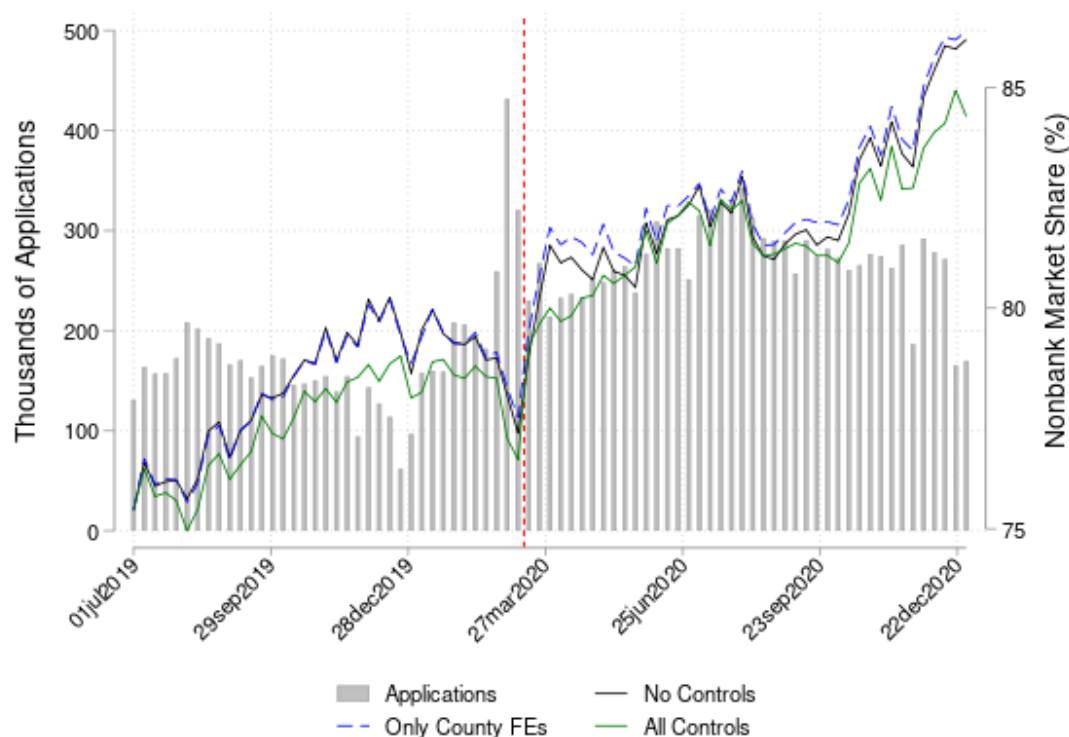


Figure IA.14. Nonbank market share: government loans. Nonbank share of government mortgage lending (e.g., FHA, VA) plotted against application date, constructed using confidential-use HMDA data. “No controls” plots the raw nonbank share of lending. “Only county FEs” plots the nonbank market share controlling for geography, estimated by regressing a nonbank dummy on time dummies and county dummies, then plotting the estimated time dummies. Similarly, “All controls” plots the nonbank share conditional on a larger set of controls, including dummies for refinancing and cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI², credit score, credit score², LTV, LTV², bins of applicant age, county dummies, and dummies for missing values of each variable. Vertical line indicates the onset of the pandemic, defined as the declaration of a national state of emergency on March 13, 2020.



V. Additional Evidence on Competition and Shopping

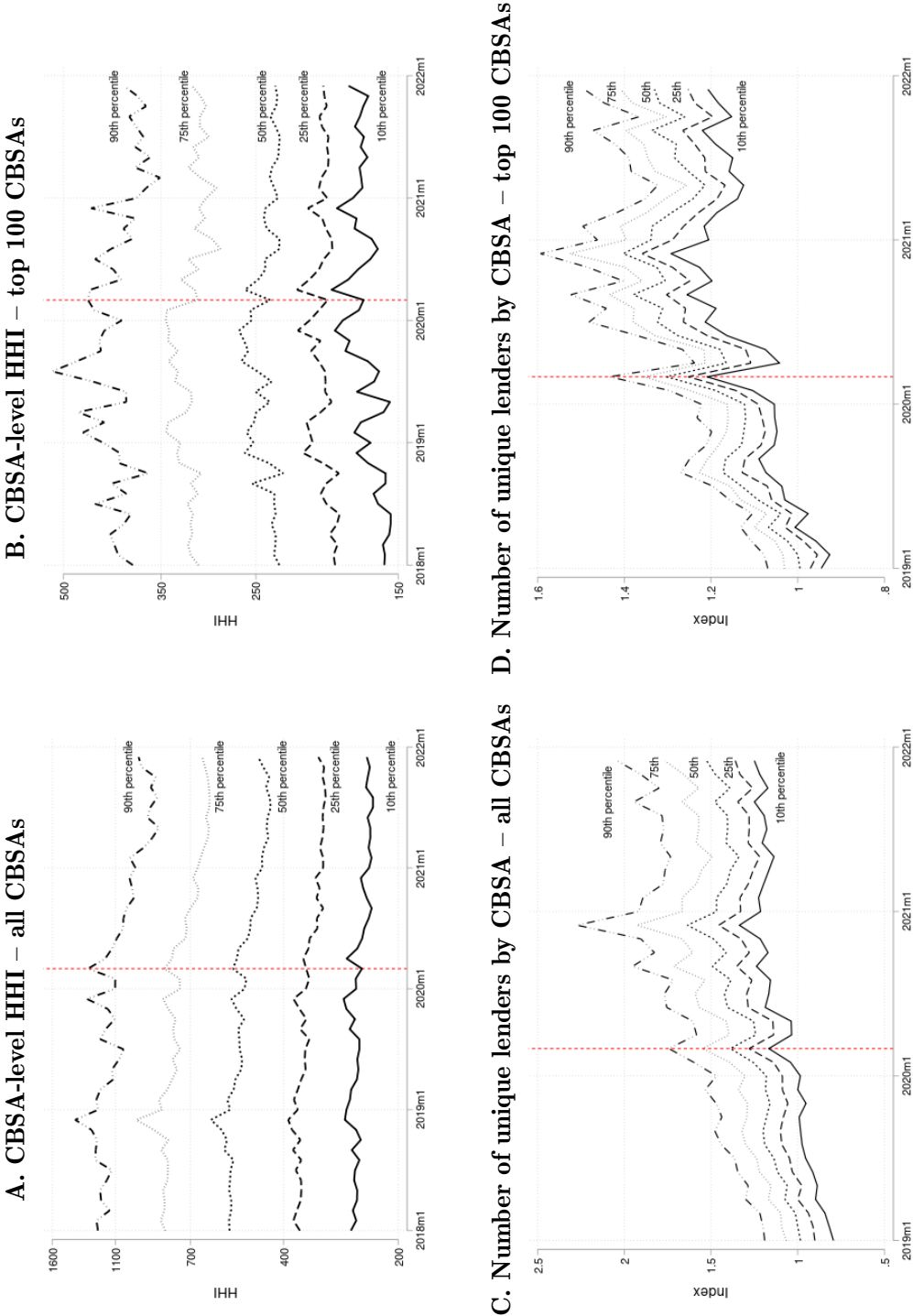
A. *Did Market Concentration Increase During the Pandemic?*

Panel A of [Figure IA.15](#) shows the distribution of monthly HHIs of originated mortgages calculated at the CBSA-application month level. The y-axis uses a log-scale. Panel B presents the same data but restricted to the top 100 most populous CBSAs. While there is considerable variation in the amount of concentration between the 10th and 90th percentile CBSA, all markets feature a large number of lenders, and there is a decline in concentration during the pandemic period.

Panels C and D plot the number of unique lenders that receive at least one application from a borrower in a given CBSA in a given month, relative to the number of unique lenders in the same CBSA in the same calendar month in 2018. (For example, the data point in Jan 2020 is the number of unique lenders in a CBSA in Jan 2020 divided by the number of unique lenders in a CBSA in Jan 2018.) Panels A and C use all 926 CBSAs in the data, while Panels B and D focus on the top 100 CBSAs by population. Both panels show that, with the exception of a short dip in April and May 2020, the number of unique lenders serving each CBSA increased during 2020, building on an existing trend of a greater number of unique lenders serving each CBSA.

To summarize, these charts show that the local market concentration of lenders tended to *decrease* during the pandemic period, when measured in terms of metro area-level Herfindahl-Hirschman Indices for lenders or the number of unique lenders serving each metro.

Figure IA.15. Evolution of market concentration and number of active lenders. These plots display the 10th, 25th, 50th, 75th, and 90th percentile values of two measures: a CBSA-by-month-level HHI calculation of lender concentration (Panels A and B, which use a log-scale on the y-axis) and the number of unique lenders (HMDA reporters) taking applications in a CBSA each month (Panels C and D). Panels A and C include all CBSAs, while B and D include the 100 most populous CBSAs. The red vertical line in each panel indicates the onset of the pandemic, defined as the declaration of a national state of emergency in March 2020. Data sources: confidential-use HMDA and Census 2018 5-Year American Community Survey.



B. Did Mortgage Shopping Behavior Change During the Pandemic?

One possible driver of higher markups during the pandemic could be a reduction in prospective borrowers' search activity, giving lenders higher effective pricing power. In this section, we test this hypothesis by examining data on borrower search and shopping from the National Survey of Mortgage Originations (NSMO) and from Google Trends. This analysis supports our discussion of lender market power and shopping in Section IV.D of the main text.

B.1. NSMO Evidence

The NSMO is a quarterly survey of borrowers with newly originated closed-end first-lien residential mortgages in the United States, undertaken by the Federal Housing Finance Agency and the Consumer Financial Protection Bureau. The nationally representative survey inquires about borrowers' experiences getting a mortgage, their perceptions of the mortgage market, and their future expectations. The survey collection began in 2014, and in most quarters, the data include between 1,000 and 1,900 responses.

We use the most recent public version of the data, released in July 2024 and covering loan originations through 2021.⁴ We focus on loans originated during the period from mid-2019 through end-2020. Since we do not know a borrower's application date, we consider loans originated from April 2020 onward as being potentially affected by the pandemic (given a typical time lag between application and origination of around two months).

To study shopping behavior, we consider six outcome variables:

1. A dummy for whether a borrower indicates that they “seriously considered” more than one mortgage lender/broker before choosing where to apply for their mort-

⁴The data are available at <https://www.fhfa.gov/nsmodata>.

gage.

2. A dummy for whether a borrower indicates that they applied to more than one mortgage lender/broker, *and* that they did so because they were searching for better loan terms.
3. An index of information use across different sources, standardized to have a mean 0 and standard deviation of 1.⁵
4. A dummy for whether a borrower indicates that having a local office or branch nearby was important in choosing the mortgage lender/broker they used.
5. A dummy for whether a borrower indicates that having a paperless online mortgage process was important in choosing the mortgage lender/broker they used.
6. A dummy for whether a borrower indicates that having used the mortgage lender/broker previously to get a mortgage was important for their choice.

The first two variables are perhaps the most “direct” measures of mortgage shopping. The third one provides a composite indicator of information search and use. The remaining three are more indirect measures, but ones that could potentially be affected by the pandemic—perhaps borrower preferences shifted toward local lenders, lenders with a strong online presence, or those that they have dealt with before. Any such shifts could indicate a change in the relative market power of affected lenders.

⁵Specifically, we sum the answers to the “x08” questions (“How much did you use each of the following sources to get information about mortgages or mortgage lenders?”), where “A lot” gets a score of 2, “A little” a score of 1, and “Not at all” a score of 0. The information sources are: your lender or mortgage broker; other mortgage lenders/brokers; real estate agents or builders; material in the mail; websites that provide information on getting a mortgage; newspaper/TV/radio; friends/relatives/co-workers; bankers, credit unions or financial planners; and housing counselors.

We regress each of these variables on a set of month dummies (with March 2020 as the omitted base category) as well as a large set of controls for borrower and loan characteristics.⁶ Results are shown in [Figure IA.16](#). As discussed below, we find no overall evidence of a decline in search intensity, and in fact on most dimensions, there is little hard evidence of significant shifts in shopping behavior during the pandemic period. Specifically:

- For the “considered multiple lenders” and “applied to multiple lenders in search of better loan terms” outcome variables, we notice a slight negative trend during the pre-COVID period; this can potentially be explained by the fact that shopping activity generally tends to decrease as rates fall, as shown by Bhutta et al. ([2024](#)). However, this downward trend stops after March 2020, and, if anything, borrowers become *more* likely again to consider or apply to multiple lenders.
- We do not see any changes in the intensity of information use.
- There is no increased preference for local lenders, except perhaps for loans originated in April 2020 (although even for this month, the coefficient estimate is not statistically significant and is lower than during two months in the 2019 pre-period). From June 2020 onward, the preference for local lenders tends to be lower than it was pre-pandemic, even though the difference is not statistically significant.
- Perhaps surprisingly, there is only weak evidence of a shift toward an increasing preference for a paperless online process. The coefficients in 2020 are higher than in 2019, although there is some evidence of a pre-trend in that values

⁶The controls are listed in the notes of [Figure IA.16](#). Note that the only control that qualitatively affects the results is the loan purpose dummy, as shopping behavior differs between purchase and refinance loans, and the share of these loan purposes shifted over our sample period.

were already higher in January and February of 2020, prior to the onset of the pandemic. That said, in a regression where the month dummies are replaced with a single COVID-period dummy equal to 1 from April 2020 onward, the coefficient on this dummy is positive with a p -value of 0.055 for the “paperless online process = important” outcome.

- Finally, borrowers’ preferences for using a lender they had previously done business with weakens over the course of 2020. Employing the same test as above, the difference between the average pre- vs post-pandemic value for “prior relationship = important” is negative, with a p -value of 0.064.⁷

To summarize, we find no evidence from this analysis of NSMO data that borrowers searched less intensively, although there is some evidence that the pandemic reduced the value borrowers put on using a lender they had previously done business with, a change that if anything should *enhance* competition. This finding is also consistent with evidence from Black Knight (2020) that “servicer retention”, that is, the share of borrowers refinancing through their existing servicer, fell to unusually low levels during the pandemic.

B.2. Google Trends

Google Trends data on borrower internet search activity provides another measure of shopping intensity. We construct a composite index of search volume by averaging searches for the terms “mortgage rate,” “mortgage refinance,” “mortgage lender,” “refinance cost,” and “mortgage cost”.

Panel A of Figure IA.17 shows that searches for these terms spiked in March 2020 to levels far above any month going back to 2012. Search activity remained elevated

⁷These last two outcomes are the only ones for which the COVID dummy is statistically significant at $p < 0.1$.

until late 2021.

Panel B tests whether borrower search activity was also unusually high relative to what would be predicted based on the level of refinance incentives, measured by the WAC-10-year spread as in our earlier analysis. We find a positive relationship between search volume and the refinancing incentive, and that borrower search activity was unusually high in 2020 and 2021 relative to what would be predicted based on the level of the refinancing incentive.

This evidence suggests that borrowers were using internet search very actively when shopping for mortgages in 2020-21, and if anything were searching more intensively during the pandemic than in prior refinancing booms.

Figure IA.16. Changes in mortgage shopping behavior/preferences over 2019-20. Each chart shows estimated calendar month coefficients and 95% confidence intervals (with March 2020 as omitted base category). Additional controls are included in each regression: loan amount category, FICO score, number of borrowers, indicators for purchase and cashout refinancing purpose, indicator for first-time homebuyers, indicators for loan program (GSE, FHA, jumbo, etc.), indicator for fixed-rate mortgage, LTV (and indicator for whether there is a junior lien), term, indicator for borrower race/ethnicity, education, age, gender, income, wealth components, risk aversion, and indicator for CRA low-to moderate income tract. Observations are weighted by the sample weights provided. Data source: NSMO.

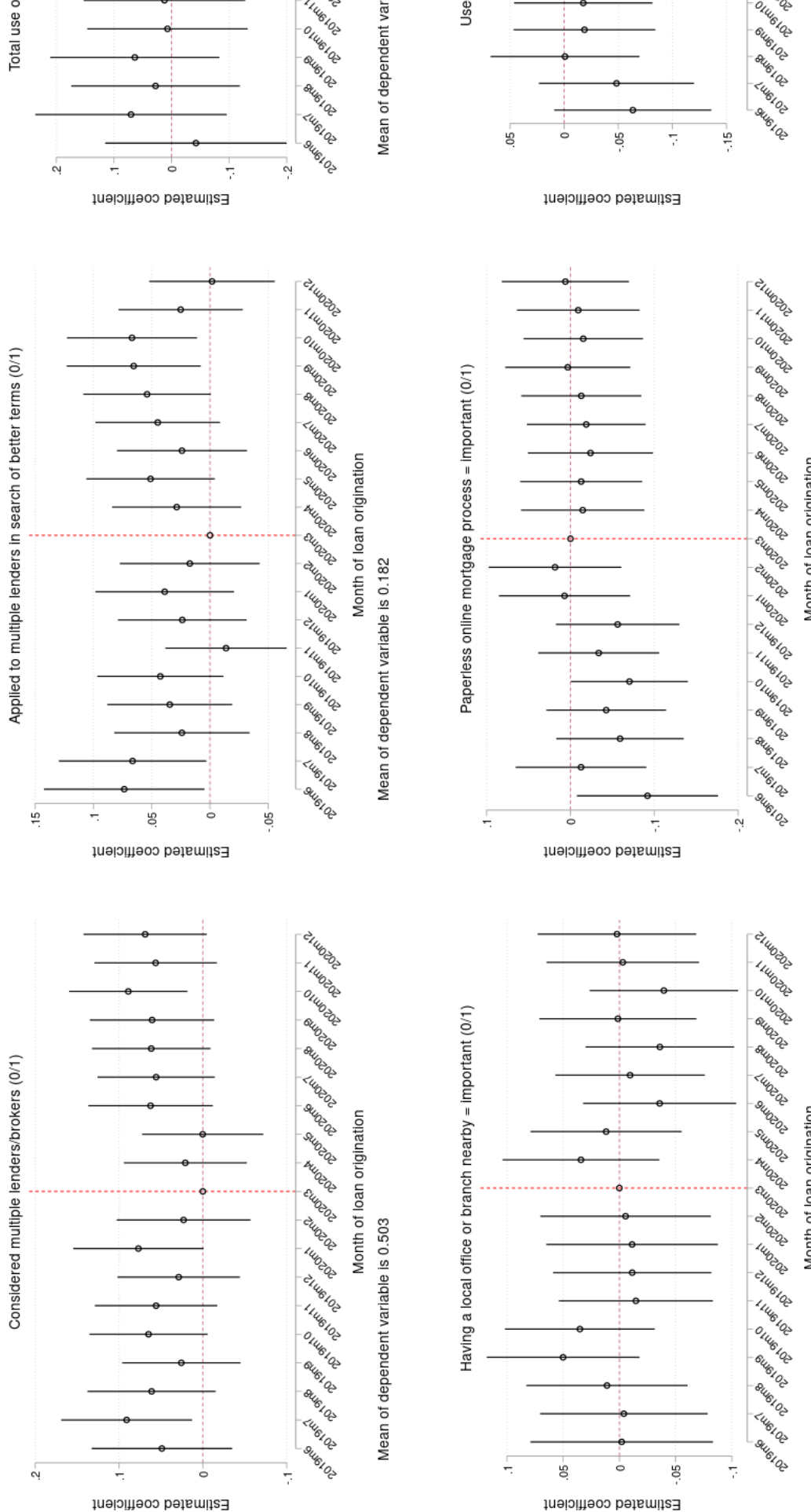
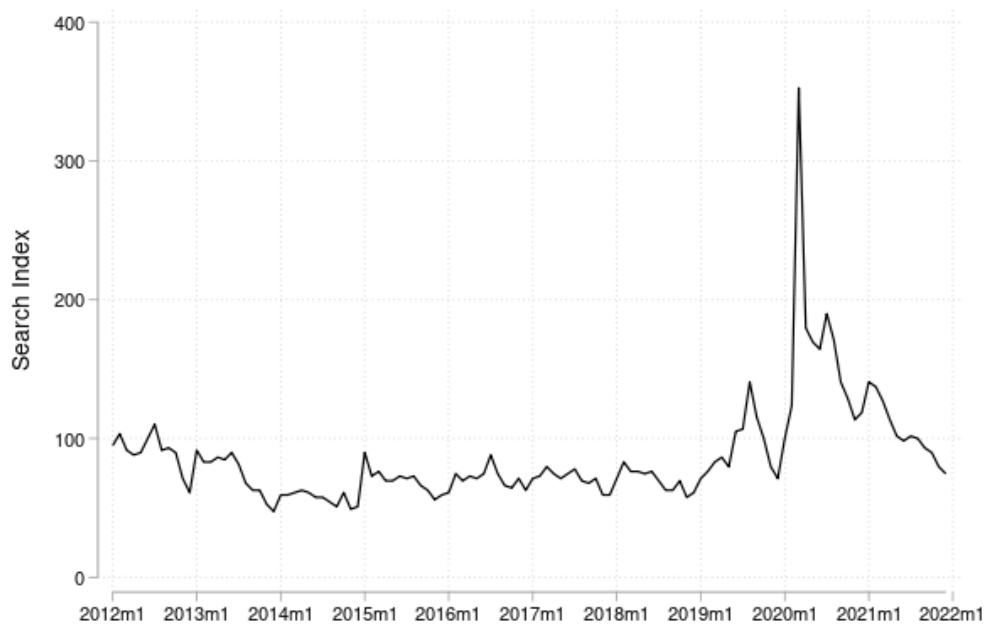
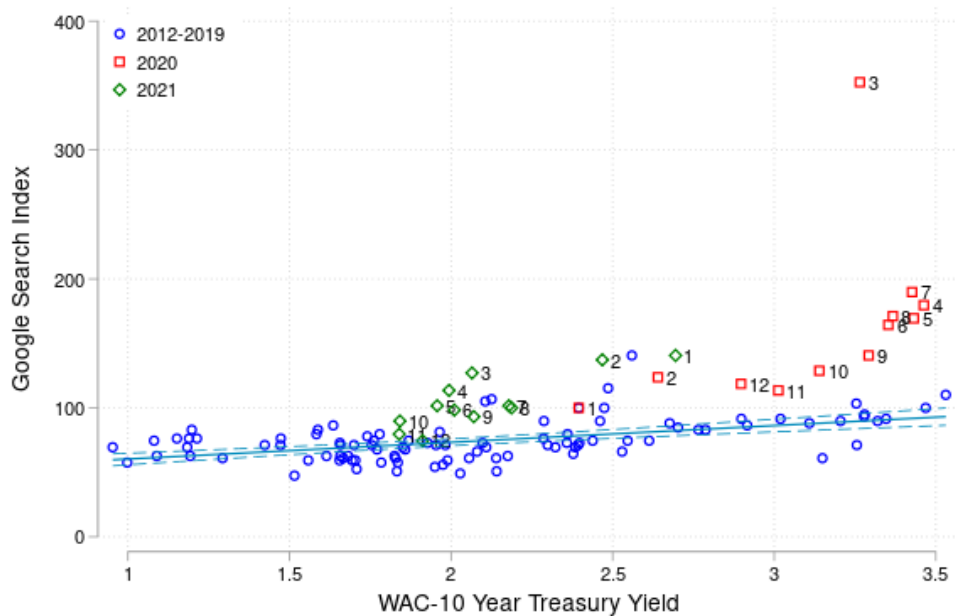


Figure IA.17. Web searches for mortgage refinancing. The Google search index is constructed by averaging the searches for the following terms: mortgage rate, mortgage refinance, mortgage lender, refinance cost, mortgage cost. Data source: Google Trends.

A. Google Trends search index for refinance related terms



B. Google searches vs. the refinance incentive



VI. Additional Descriptive Statistics

A. *Mortgage Non-payment by Credit Score Bin*

Figure IA.18 below presents a binscatter plot of the proportion of mortgages that were 60-plus days past due as of January 2020 and June 2020. The figure is based on McDash servicing data for mortgages originated since 2017. FHA loans are flagged based on the static loan type field in McDash. Figure IA.19 shows the change in the past-due rate from January to June.

Non-payment rates increased across the credit score distribution for GSE, FHA, and portfolio loans. However, the rise in past-due rates was more pronounced for low-credit-score loans, consistent with the expectation that the pandemic has particularly amplified the credit risk for loans to less creditworthy borrowers. Note that a large share of these borrowers, although not all, took advantage of forbearance programs (e.g., Kim et al., 2024; An et al., 2021).

Figure IA.18. Non-payment vs. credit score: pre-pandemic vs. pandemic. Data source: McDash data from ICE. Note: FHA loans are all classified as FHA, regardless of ownership. Conventional loans are categorized as GSE or portfolio based on the ownership status of the loan at the time of the data snapshot. "All loans" includes FHA, GSE, and conventional loans in portfolio, as well as VA loans and conventional loans held in private-label securities.

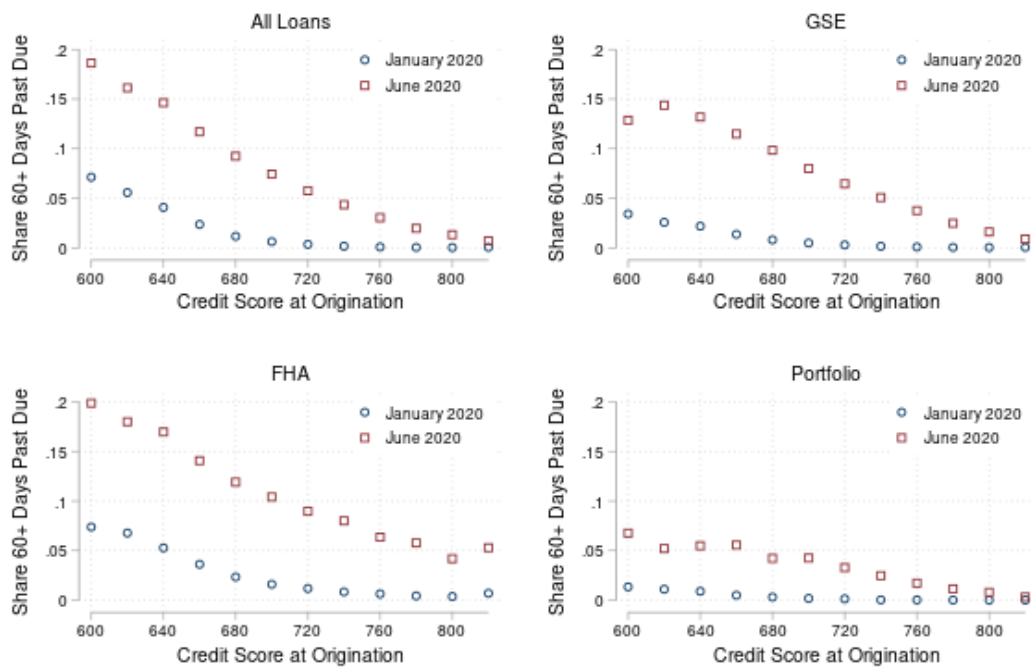
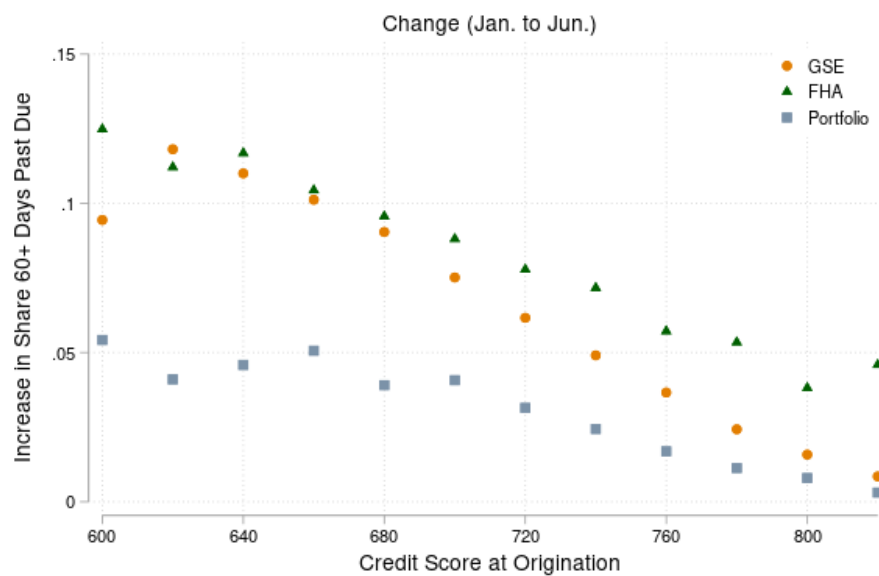


Figure IA.19. Non-payment vs. credit score: pre-pandemic vs. pandemic (percentage point change). Data source: McDash data from ICE. FHA loans are all classified as FHA, regardless of ownership. Conventional loans are categorized as GSE or portfolio based on the ownership status of the loan at the time of the data snapshot.



B. Credit Score Distributions by Market Segment

Figure IA.20. Credit score distribution: conventional conforming loans. Data source: confidential-use HMDA data. Includes originations of 30-year, fixed-rate conventional mortgages made to owner-occupants, weighted by loan balance at origination. Loans are grouped by application date weekly on Thursdays, including loans since the prior Friday. Dashed vertical line is March 2020.

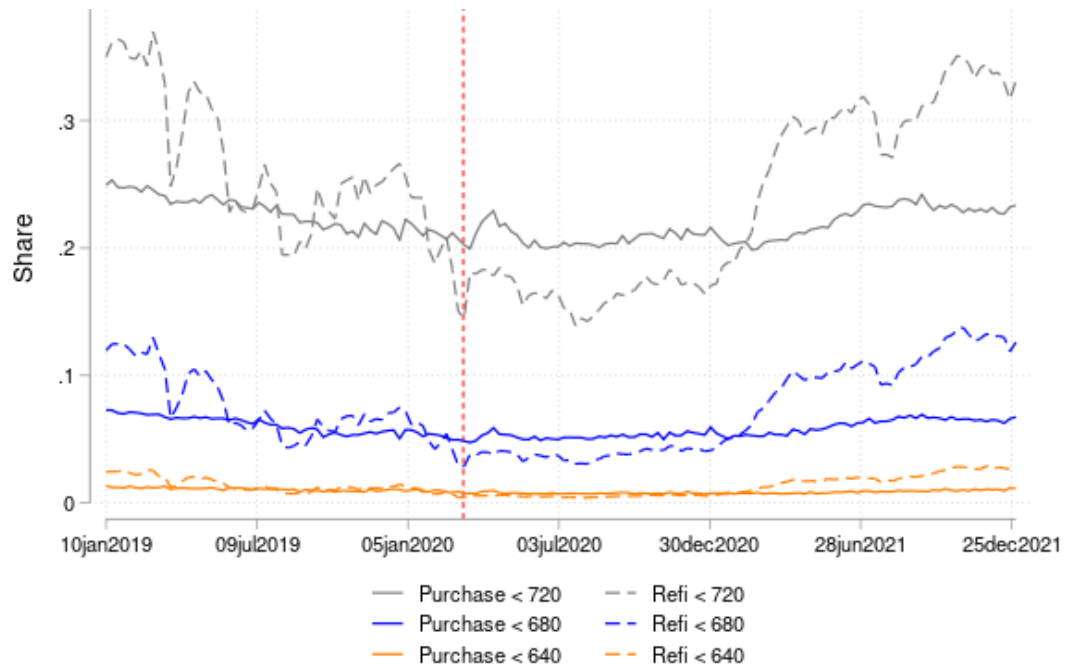


Figure IA.21. Credit score distribution: FHA loans. Data source: confidential-use HMDA data. Includes originations of 30-year, fixed-rate FHA mortgages made to owner-occupants, weighted by loan balance at origination. Loans are grouped by application date weekly on Thursdays, including loans since the prior Friday. Dashed vertical line is March 2020.

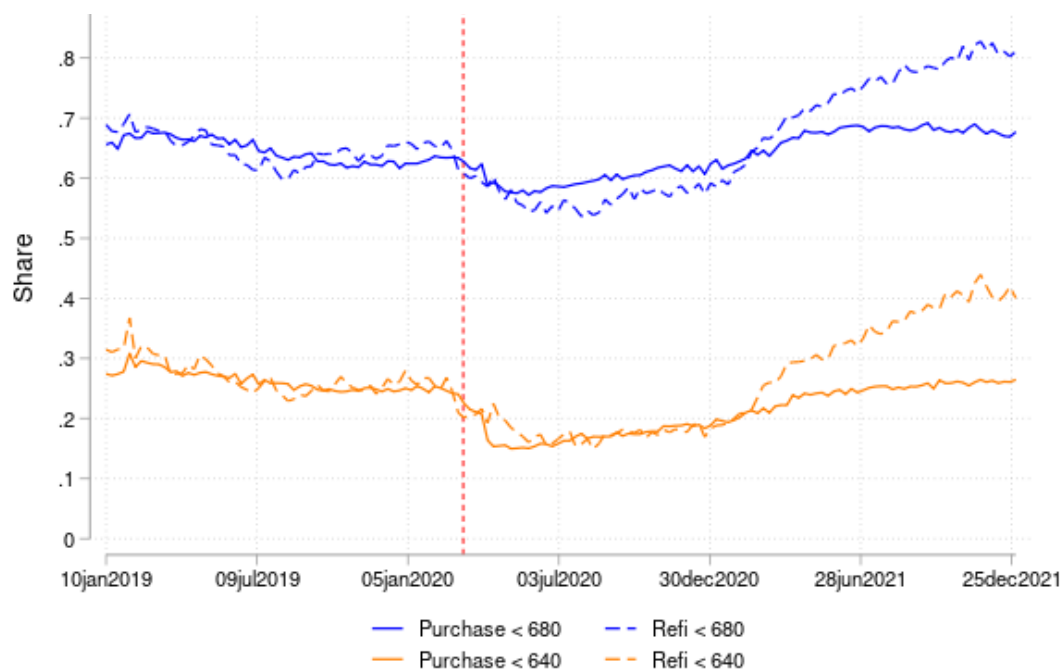
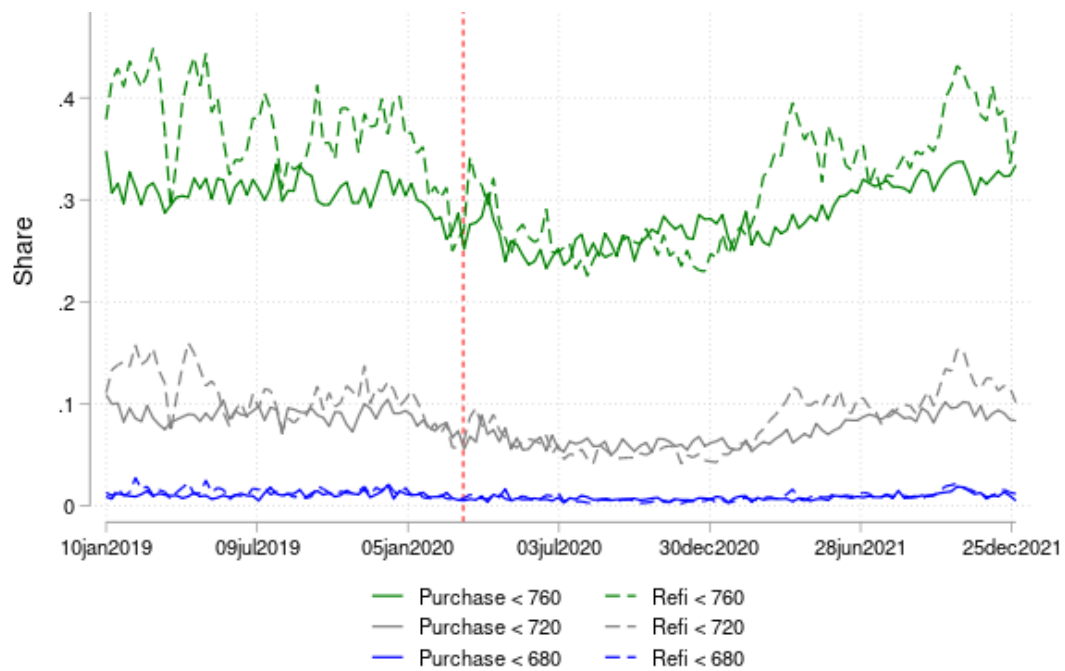


Figure IA.22. Credit score distribution: jumbo loans. Data source: confidential-use HMDA data. Includes originations of 30-year, fixed-rate conventional mortgages made to owner-occupants, weighted by loan balance at origination. Loans are grouped by application date weekly on Thursdays, including loans since the prior Friday. Dashed vertical line is March 2020.



VII. Analysis of Credit Supply in Riskier Segments

This section analyzes credit supply in two riskier market segments, the jumbo market and the FHA market. The empirical results presented below underlie the discussion of credit supply in these two segments presented in Section V of the main text. We also provide additional institutional details (e.g., discussion of the sources of risk facing intermediaries in these two segments).

A. *The Jumbo Market*

Jumbo mortgages are large loans exceeding the conforming limits for agency securitization. Jumbo borrowers typically have high incomes and credit scores, but jumbos are riskier for lenders and investors (usually banks) because they do not carry a government guarantee against credit risk. Comparing credit conditions in the jumbo and conforming markets therefore sheds light on whether these guarantees stabilized credit supply in the pandemic.

As part of our analysis, we also examine the supply of “superconforming” (or “conforming jumbo”) loans. These are mortgages which exceed the national conforming loan limit but are still eligible for agency securitization because they are located in a county with a higher local limit.⁸ Superconforming loans are still eligible for government guarantees, but for institutional reasons they are somewhat less liquid and are also less likely to be purchased by the Federal Reserve in its quantitative easing (QE) program. Studying the supply of superconforming loans after the resumption of Fed

⁸Fannie Mae and Freddie Mac cannot purchase or securitize mortgages exceeding the relevant conforming loan limit. The national conforming limit (\$510,400 in 2020 for a single-family home) applies in most counties, but the limit is higher in counties with high home prices, up to \$765,600 in 2020. A superconforming mortgage is a loan with a principal balance between the national limit and the relevant local limit. See Vickery and Wright (2013) for further institutional details.

QE in March 2020 can shed light on how Fed MBS purchases affected mortgage credit supply, and in particular whether QE disproportionately boosted supply for the loans most likely to be purchased by the Fed.

A.1. Evidence from Optimal Blue

To begin, we use Optimal Blue data to estimate the interest rate spread on jumbo mortgages compared to smaller but otherwise identical conforming mortgages, following our earlier methodology. Results are presented in [Figure IA.23](#).

Panel A plots the jumbo-conforming offer rate spread for a prime mortgage with LTV of 80 and credit score of 750. Before the pandemic, this spread was fairly stable at 10-25bp. During the pandemic, however, it increases sharply to 80-100bp from April through August 2020, before partially normalizing to a level of about 50bp by the end of 2020.⁹ The jumbo-conforming spread based on rate locks (Panel B) follows a similar pattern; the spread rises from 0-10bp to a peak of 40-50bp from April through August, then declines to about 30bp by December. (Note: These rate lock estimates are based on the same regression model approach we used to study the conforming market in [Section IV.B](#) of the main text.) We also see a similar rise and fall in the jumbo-conforming spread in time-series rate data from *Mortgage News Daily* and the Mortgage Bankers Association—see [Figure IA.24](#).

We also use rate lock data to track the evolution of the superconforming-conforming spread. These estimates are reported in Panel B of [Figure IA.23](#), based on the same methodology (i.e., we include a vector of superconforming dummy $\times$ time variables in our rate lock model, then plot the coefficients and confidence intervals). The superconforming spread does indeed rise early in the pandemic, from 10-20bp to 30-35bp,

⁹Optimal Blue data also indicate a large increase in the jumbo-conforming spread for 5/1 adjustable-rate mortgages.

coincident with the resumption of Fed MBS QE in mid-March 2020, suggesting that QE had “local supply” effects in the conforming market where purchases were concentrated. This increase is less persistent than the rise in the jumbo-conforming spread, however. See Section A.3 for further discussion.

Turning to quantities, we observe a sharp contraction in the number of lenders offering jumbo mortgages during the pandemic, particularly for riskier borrowers (Panel C of Figure IA.23). The number of lenders offering jumbo loans to prime borrowers with a credit score of 750 drops by more than half in the early stages of the pandemic, before slowly recovering to about 20% below pre-pandemic levels by December 2020. For lower FICO borrowers (680 and 640), the number of active lenders collapses—quite strikingly—almost to zero, and remains low through year-end. Finally, Panel D of Figure IA.23 analyzes the share of rate locks that are for jumbo loans rather than conforming loans. This share declines by about half, both for purchase loans and refinances, before recovering.

Figure IA.23. Credit supply in the jumbo market. Measures of credit supply computed using the methodology described in Section IV.B of the main text. Vertical line represents the declaration of a national state of emergency on March 13, 2020. Data source: Optimal Blue.

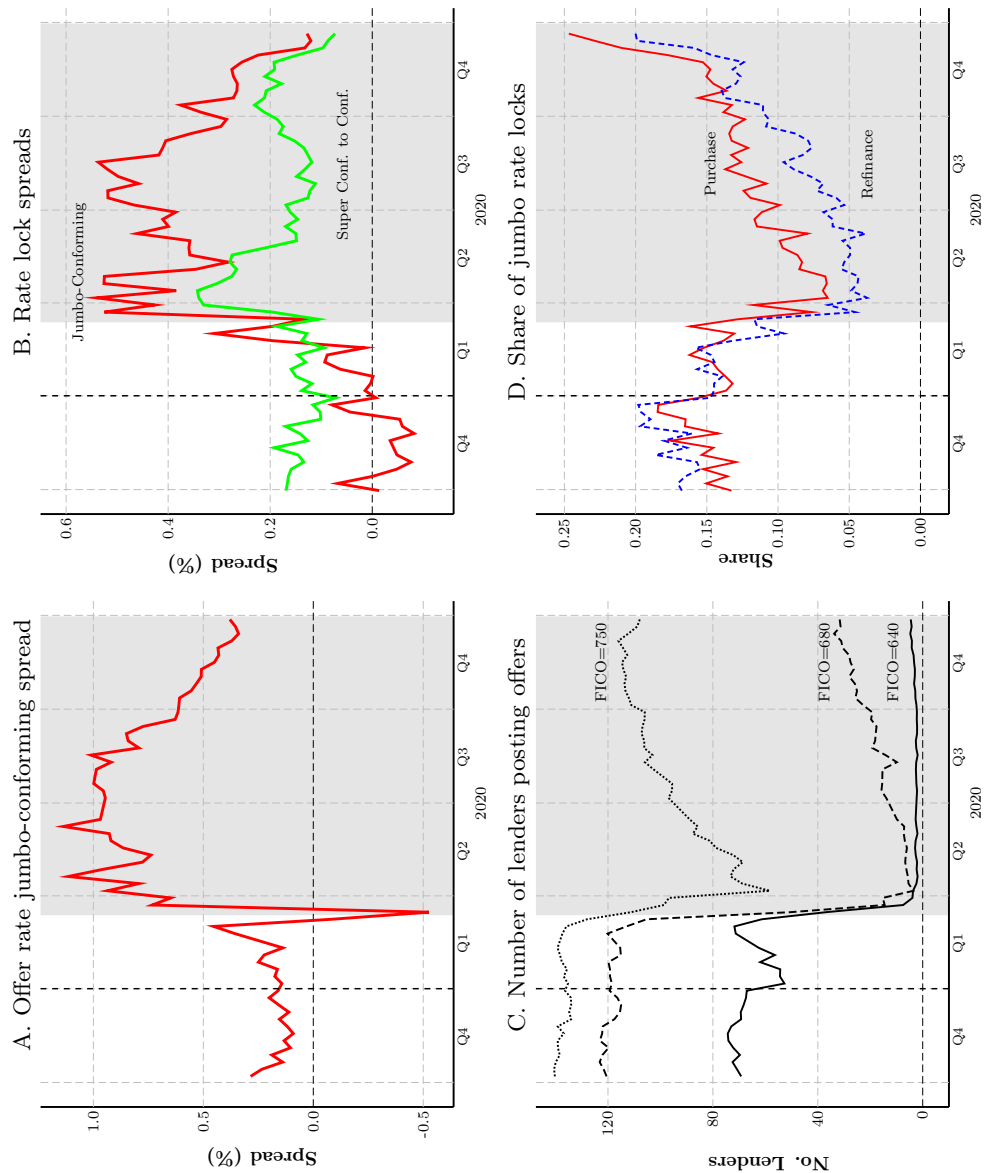
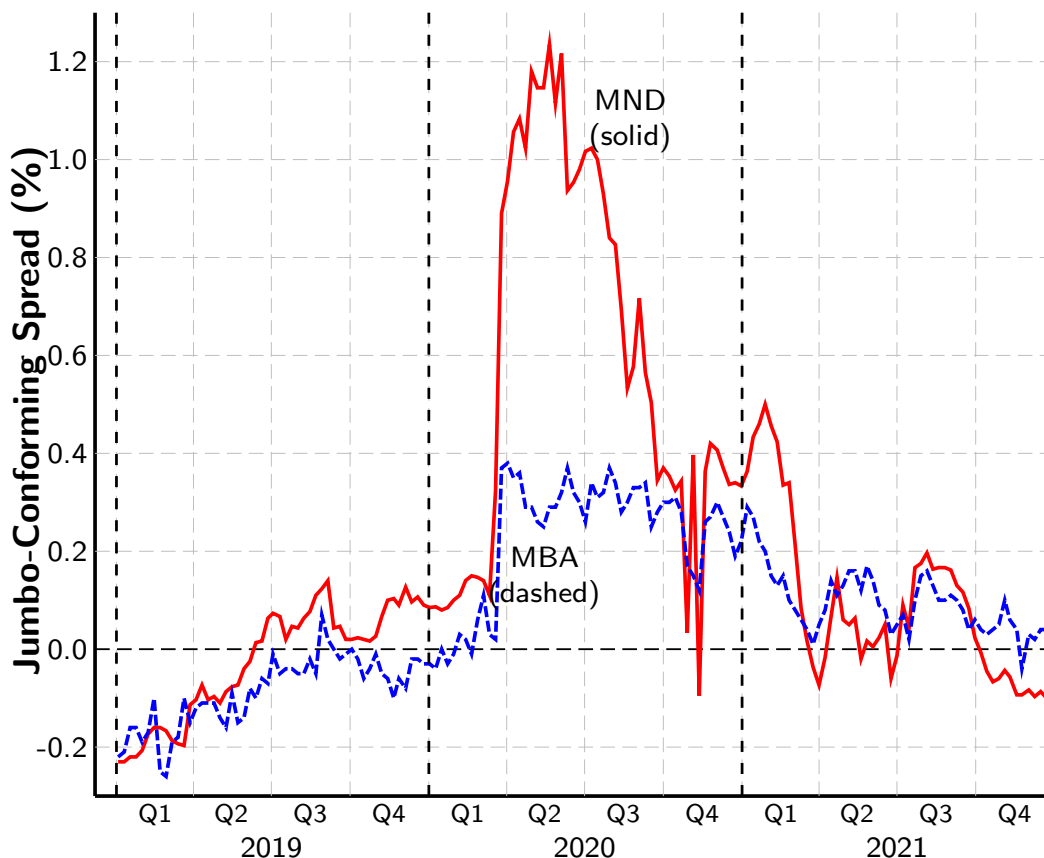


Figure IA.24. Jumbo-conforming spreads from other data sources. Vertical line represents the declaration of a national state of emergency on March 13, 2020. Data sources: *Mortgage News Daily* and Mortgage Bankers Association (via Haver Analytics)



A.2. Evidence from HMDA

This evidence on rates and quantities indicates a significant negative supply response in the jumbo market, although an important caveat is that Optimal Blue primarily reflects mortgages originated by nonbanks. Banks retain a strong presence in the jumbo market, and several large banks stopped purchasing jumbos from nonbanks during the pandemic (Eisen, 2020). A potential concern, therefore, is that the drop in quantity we observe in Figure IA.23 simply reflects substitution in lending from nonbanks to banks.

For a more representative picture of the jumbo market, we turn to loan-level

confidential-use HMDA data, which have excellent coverage of both banks and non-banks. We estimate linear probability models where the dependent variable equals 100 for a jumbo loan and zero otherwise. The key explanatory variable is a pandemic dummy, defined as an application date in March 2020 or later. In some specifications we control for a comprehensive set of loan and borrower controls to soak up changes in the composition of lending. We focus on a narrow window of 10% around the applicable conforming loan limit (either the national limit, or the relevant higher local limit for mortgages in high-cost counties).

Regression results presented in [Table IA.IV](#) show that there was indeed a significant decline in the fraction of jumbo loans, of about 7-9 percentage points. These results are fairly similar in magnitude across purchase mortgages and refinances and are also robust to whether or not we control for loan characteristics. (Note: the set of controls used is listed in the table notes.)

To zoom in further, [Figure IA.25](#) traces out the dynamic evolution of the jumbo share over this period, by reestimating the model from [Table IA.IV](#) replacing the pandemic dummy with a vector of application month dummies, then plotting the estimates and confidence intervals on these dummies. In 2019, shortly before the pandemic, jumbo loans make up about 20% of purchase volume and 17-18% of refinancing volume in the segment of the loan size distribution around the conforming limit. But the jumbo share declines sharply in 2020, reaching lows of about 8% for purchase loans and 7% for refinances by summer 2020, less than half of the pre-pandemic levels. We then see a partial recovery later in 2020, consistent with the decline in the jumbo rate spread documented earlier.

Figure IA.25. Evolution of jumbo lending share. This figure displays estimated coefficients and associated 95% confidence intervals on a vector of time dummies from linear probability models where the dependent variable = 1 if the mortgage is a jumbo loan. The sample includes loans within 10% either side of the conforming loan limit. See notes from [Table IA.IV](#) for a list of controls and other details. Models estimated using confidential-use HMDA data.

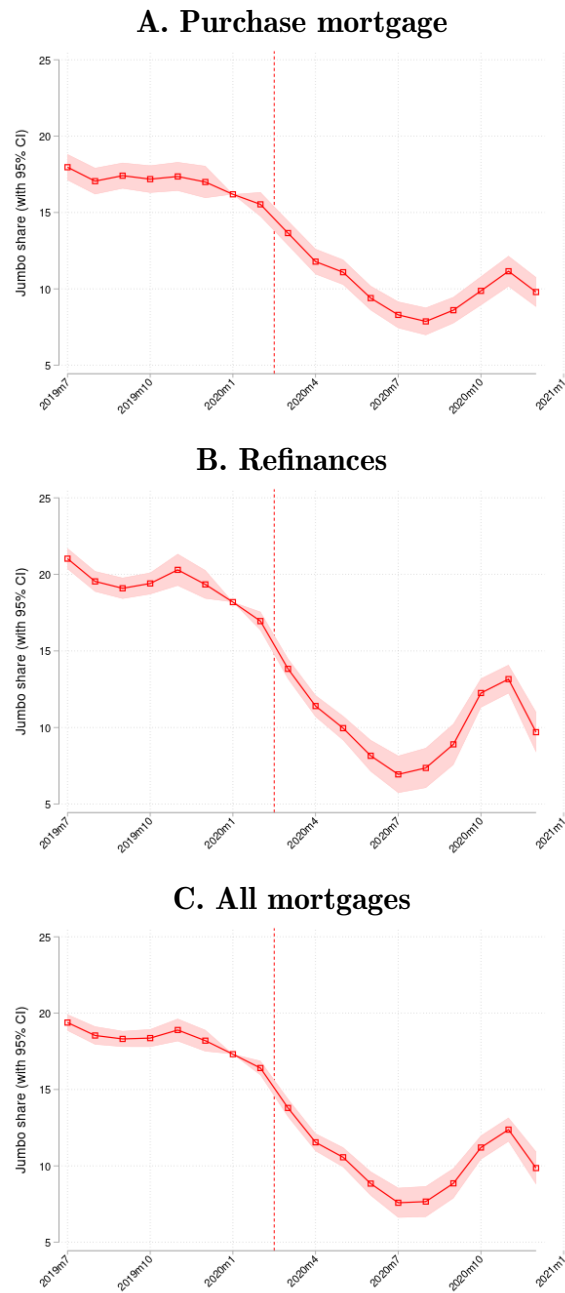


Table IA.IV. Change in Jumbo Share during the Pandemic. Linear probability model, estimated using loan-level confidential-use HMDA data. Sample includes loans within 10% either side of the conforming loan limit (CLL) applicable to each loan (either the national or county-level limit, whichever is applicable). Pandemic dummy = 1 if application date is March 2020 or later. Loan controls include: county dummies, debt-to-income (DTI), DTI², loan-to-value (LTV), LTV², credit score, credit score², log(appraisal amount), log(appraisal amount)², log(applicant income), log(applicant income)², applicant sex, applicant race and ethnicity, applicant age, an indicator for having no co-borrower, occupancy status, binned applicant age, and loan purpose (purchase, rate/term refinance, or cash-out refinance), where applicable. Sample period is July 2019 to December 2020. Robust standard errors in parentheses clustered at the county level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Pandemic	-7.197*** (0.215)	-7.158*** (0.217)	-8.075*** (0.390)	-8.989*** (0.408)	-7.657*** (0.289)	-8.098*** (0.312)
Num obs.	397861	397396	511947	511573	909808	909328
Mean of dep. var.	12.48	12.47	13.18	13.18	12.87	12.87
Origination type	Purchase	Purchase	Refinance	Refinance	All	All
Loan Controls	N	Y	N	Y	N	Y
County FE	N	Y	N	Y	N	Y

A.3. Disentangling Credit Guarantees vs. QE

An additional factor affecting the relative supply of jumbo and conforming loans in 2020 was the rapid expansion of Federal Reserve holdings of conforming mortgages through its agency MBS portfolio. The Fed purchased \$580bn in agency MBS through the TBA forward market in March and April 2020 alone (Frame et al., 2021), and its agency MBS holdings increased rapidly from \$1.37 trillion at the start of March 2020 to \$1.90 trillion by the end of June (source: Federal Reserve Bank of New York).¹⁰

¹⁰Note that there is some delay between the timing of the Fed purchases and the growth in the portfolio because the TBA forward trades settle only once per month. The Fed also executed a significant volume of dollar roll transactions during this period to delay settlement of its purchases. See Frame et al. (2021) for further discussion.

The pace of Fed TBA purchases slowed after April 2020, but the Fed continued to accumulate securities through 2020-21, with the portfolio peaking in size at \$2.7tr in early 2022.

This is important because research on quantitative easing (QE) shows that central bank asset purchases have “local” effects on the markets where purchases are directly concentrated (e.g., Krishnamurthy and Vissing-Jorgensen, 2011). Therefore, this rapid expansion in the Fed’s portfolio would be expected to disproportionately affect credit supply in the conforming market *relative* to the jumbo market, where the Fed was not an active buyer.

Time-series evidence suggests QE did have a significant effect on credit conditions. First, Figure 2 in the main text shows that MBS yields declined sharply just after QE was announced, including a decline in OAS by about 40bp, and a fall in option cost due to a drop in interest rate volatility. (These changes were almost exactly offset by a rise in the primary-secondary spread, however, leaving the mortgage-Treasury spread little changed.) Second, the resumption of QE was associated with an increase in the jumbo-conforming spread in Figure IA.23, indicating that asset purchases had disproportionate effects on the conforming market.

A limitation of these tests, however, is that the resumption of QE is closely tied to the financial and economic pressures at the time, which also likely amplified credit risk premia in the jumbo market. We therefore turn to the superconforming market as arguably a cleaner way to disentangle the supply effects of QE and credit guarantees.

Superconforming mortgages are useful for identification because they are guaranteed, but are significantly less likely to be purchased by the Fed. This is because the Fed obtains agency MBS pools through the “to-be-announced,” or TBA, market, and pools comprising more than 10% of superconforming loans are not TBA eligible (Vickery and Wright, 2013; Huh and Kim, 2021). Matching eMBS loan-level data to

security-level data on the Fed’s MBS holdings, we confirm that the probability of an agency mortgage ending up in a pool in the Fed’s MBS portfolio does indeed drop sharply just above the national conforming limit, declining by about 40%.

As we showed in Panel B of [Figure IA.23](#), the superconforming-conforming spread does indeed spike upwards just after the resumption of Fed MBS QE in mid-March, by about 20bp. The *jumbo*-conforming spread also rises, by a larger amount—about 35-40bp. This suggests that both QE and credit risk premia were important during this early phase of the pandemic (since the jumbo-conforming spread reflects both factors while the superconforming spread just reflects the former). The rise in the superconforming spread is less persistent though, lasting for only two months or so until the start of June. This matches the timing of Fed purchases—as we have discussed, the pace of forward asset purchases was much more rapid in March and April 2020 (reflecting settlements through May) than later in 2020.

Next, we use HMDA data to study effects on the quantity of credit. To separate the effects of QE from credit guarantees, we restrict the HMDA sample to mortgages in high-cost counties and we study shifts in lending in a 10% window around both the *national* conforming loan limit and the higher *local* limit. In the former case, the outcome variable is a dummy for whether the loan is a superconforming mortgage; in the latter, it is a dummy for whether the mortgage is a jumbo loan. In all specifications, we include a battery of loan controls to account for compositional shifts in lending (see table notes for a list of controls).

Results are reported in [Table IA.V](#). In columns 1-3, the outcome variable is equal to 100 for a superconforming mortgage. Focusing on column 3, the share of superconforming loans in this part of the loan size distribution drops by 7pp after the pandemic begins, compared to a sample mean of 28%. Around the higher local limit, the share of jumbo mortgages drops by 10 percentage points compared to a sample mean of 17%.

Table IA.V. Change in Jumbo and Superconforming Share, High-Cost Areas Only.

Linear probability model, estimated using loan-level HMDA data. Sample includes loans within 10% either side of the conforming loan limit (CLL). Sample is restricted to “high-cost” counties where the county-level CLL exceeds the national CLL. Pandemic dummy = 1 if application date is March 2020 or later. Loan controls include: county dummies, debt-to-income (DTI), DTI², loan-to-value (LTV), LTV², credit score, credit score², log(appraisal amount), log(appraisal amount)², log(applicant income), log(applicant income)², applicant sex, applicant race and ethnicity, applicant age, an indicator for having no co-borrower, occupancy status, binned applicant age, and loan purpose (purchase, rate/term refinance, or cash-out refinance), where applicable. Sample period is July 2019 to December 2020. Robust standard errors in parentheses clustered at the county level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	> National CLL			> Local CLL		
	(1)	(2)	(3)	(4)	(5)	(6)
Pandemic	-4.965*** (0.351)	-7.322*** (0.266)	-6.839*** (0.255)	-8.611*** (0.302)	-11.42*** (0.594)	-10.27*** (0.462)
Num obs.	201296	550549	751852	131652	210040	341702
Mean of dep. var.	33.78	26.54	28.48	16.64	17.46	17.14
Mean Y [Pre-Pandemic]	36.66	29.56	31.87	22.25	24.23	23.43
Origination type	Purchase	Refinance	All	Purchase	Refinance	All
Loan Controls	Y	Y	Y	Y	Y	Y

Therefore, the results indicate that both QE and guarantees promoted credit supply, although the effects around the local limit (the upper bound to qualify for government guarantees) are quantitatively larger, suggesting that credit guarantees had a relatively larger effect in bolstering credit supply.

We then explore the dynamics of these effects by replacing the pandemic dummy with a vector of time dummies (again, indexed by application date) and plotting them in [Figure IA.26](#) as event study plots. These estimates show that loan volume effects around the national limit (which reflect only QE) are significant in the early stages of the pandemic, but they are less persistent than the effects at the local limit (above which loans do not qualify for government guarantees). This timing is consistent with our earlier interest rate estimates in [Figure IA.23](#), which show that

the superconforming-conforming spread is elevated only through June 2020, while the jumbo-conforming spread remains high throughout 2020.

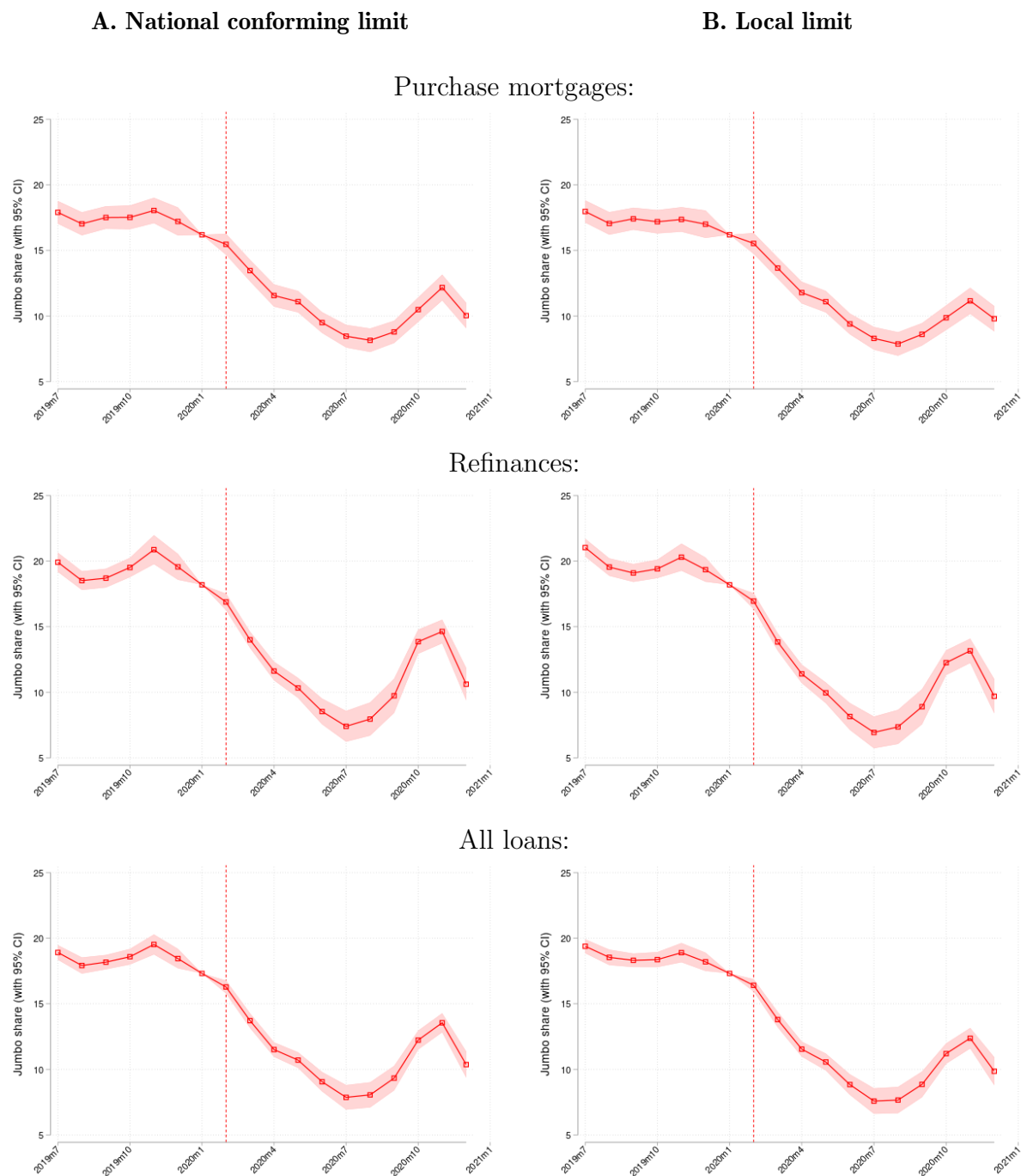
A.4. Jumbo Analysis: Summary

We find a persistent increase in interest rate spreads and a decline in quantities in 2020 for jumbo mortgages that do not qualify for government credit guarantees. Analyzing the superconforming market, we estimate that these effects are in part due to Fed QE purchases of agency MBS backed by conforming loans, particularly in the spring of 2020. We interpret the remainder of the effect as being due to an amplification of the benefits of government guarantees due to the uncertain economic environment.

An alternative explanation is that our results reflect an amplification of the *liquidity* benefits of conforming mortgages, given that conforming loans can be easily sold into a liquid secondary market while jumbos are typically held on balance sheet by banks. This “liquidity” channel seems less plausible, however, given that banks had ample liquidity in 2020 (Li et al., 2020) and at the time were rapidly expanding their holdings of mortgage-related assets. The “credit risk” explanation is also consistent with the fact that jumbo credit supply declined more rapidly for less creditworthy loans.

Finally we note that, while we find that the effects of MBS QE faded after a few months, our estimates do not reflect the full general-equilibrium effects of QE. For example, the effects of QE on credit supply may have become more diffuse over time as investors had time to adjust the composition of their portfolios.

Figure IA.26. Lending above national vs. local conforming limit. This figure displays estimated coefficients and associated 95% confidence intervals on a vector of time dummies from linear probability models where the dependent variable = 1 if the mortgage is larger than either the national or local conforming limit. The sample includes loans within 10% either side of the relevant conforming loan limit. See notes from [Table IA.V](#) for a list of controls and other details.



B. The FHA Market

The FHA market is an important segment of the mortgage market that disproportionately serves lower-income borrowers and first-time homebuyers. While FHA loans are securitized and carry a government credit guarantee, these loans present more risk for mortgage intermediaries than a prime conforming loan due to institutional factors (Kim et al., 2018). First, intermediaries face liquidity risk because when an FHA borrower defaults, servicers are typically obligated to advance payments to investors until termination or modification (compared to four months in the conforming market), and they face significant delays before being reimbursed. This additional liquidity risk is likely to be particularly important for the nonbanks that now dominate FHA lending and servicing. Second, the servicer is not generally reimbursed for all the costs associated with foreclosure, and also faces some risk that the claim will be denied entirely. Tozer (2019) estimates that uncompensated costs average about \$10,000 per FHA claim. Third, the likelihood of default is significantly higher because FHA borrowers are typically lower-income and often first-time home buyers.¹¹

These factors are important because default risk was amplified during the pandemic, in part due to the creation of forbearance programs that allowed borrowers to pause their payments (see Section IV.B of the main text for further discussion). It seems plausible that this heightened risk had a chilling effect on credit supply in the FHA market, despite the presence of government guarantees.¹² We test this “risk pre-

¹¹In June 2020, 15.7% of FHA loans were 60 or more days past due, compared with only 6.7% of conventional loans. (Source: 2020:Q2 MBA delinquency survey.)

¹²Policymakers eventually took steps to limit the risks facing FHA lenders, in particular: (1) to limit liquidity outflows, the FHA determined that loans that re-perform after exiting forbearance can be made current by issuing a partial claim, reimbursing the servicer for principal and interest advances during forbearance; (2) Ginnie Mae created a temporary liquidity facility for servicers, the Pass-Through Assistance Program, or “PTAP.”

mium” hypothesis in two ways: i) by studying interest rates and quantities *within* the FHA market, comparing loans with different levels of default risk, and ii) by comparing credit conditions in the FHA market to the conventional conforming market.

B.1. Default Risk and Credit Supply Within the FHA Market

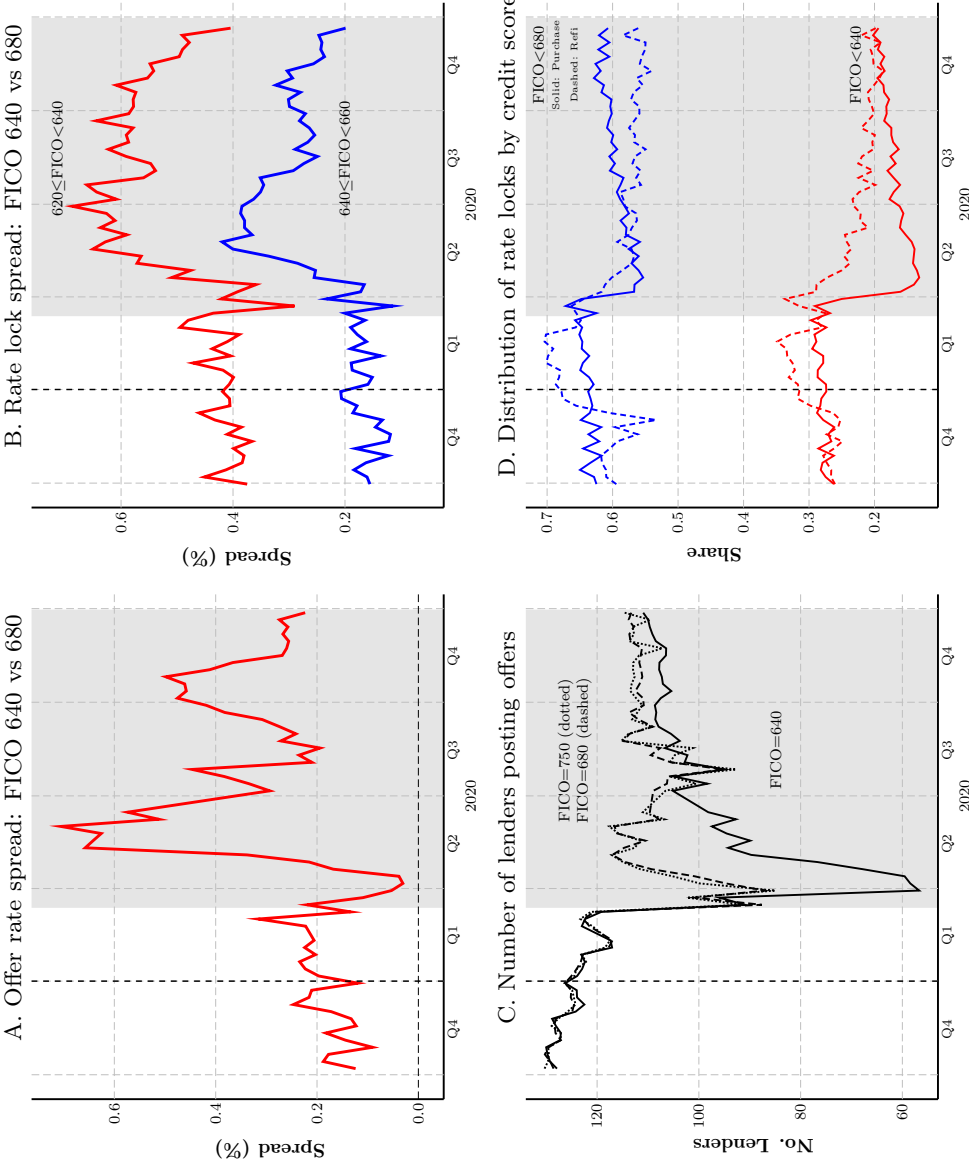
First, we use Optimal Blue data to examine rates and quantities for FHA borrowers with higher versus lower default risk (based on credit score), using the methodology described in Section IV.B of the main text. Results are presented in Figure IA.27.

Panels A and B analyze interest rates on mortgage offers and rate locks, comparing borrowers with a FICO credit score of 640 versus 680.¹³ The results show a clear increase in the interest-rate premium for the riskiest FHA borrowers. Panel A finds that in the mortgage offer data, the FICO 640-680 interest rate spread was stable around 20bp before the pandemic but rose sharply beginning in April 2020, coincident with the post-CARES Act surge in forbearance, peaking at about 70bp in June (50bp above pre-pandemic levels). The spread then decreased and displayed some volatility, but it remained elevated through the end of 2020. The low-FICO spread also increased in the locks data (Panel B of Figure IA.27), although the peak increase was only about 25bp, half as large as in the offer data.¹⁴

¹³A credit score of 680 is typical for FHA loans (roughly at the 60th percentile in 2020:Q1), while 640 is roughly at the 20th percentile; see Figure IA.21 which displays the credit score distribution of FHA originations over the 2019–2020 period. (Figure IA.20 and Figure IA.22 plot credit score distributions for the conventional conforming and jumbo markets, for comparison). Other assumptions we apply for the offer data are that we study 30-year purchase-money FRMs with zero points, LTV = 95 to 97% (which is by far the most common in the FHA segment), and DTI of 36%.

¹⁴This difference between offers and equilibrium outcomes may reflect heterogeneity in supply responses across lenders—e.g., a subset of risk-averse lenders post very high rates for low-FICO loans, but these lenders comprise only a small share of locks because their uncompetitive rates mean that few borrowers choose them. We see a similar pattern in the jumbo results in Section VII.A.

Figure IA.27. Credit supply in the FHA market. Measures of credit supply computed using the methodology described in Section IV.B of the main text. Vertical line represents the declaration of a national state of emergency on March 13, 2020. Data source: Optimal Blue.



Panels C and D of [Figure IA.27](#) study quantities, which also indicate a decline in credit supply to the highest-risk FHA borrowers. Panel C shows that the number of lenders offering any type of FHA loan drops by one-quarter during the market volatility in March 2020. In the highest-risk segment (FICO 640), the number of lenders then falls further in April, to half the pre-pandemic level. In contrast, lenders re-enter the market for lower-risk loans (FICO of 680 or 750) over the sample period. Lenders gradually return to the FICO 640 segment later in 2020. Similarly, Panel D shows that the share of FHA purchase rate locks to borrowers with a FICO score lower than 640 drops from 30% pre-pandemic to less than 15% in April, coincident with the spike in rate spreads and drop in the number of lenders.¹⁵ Notably, we do not see such a sharp decline for FHA refinances. For an FHA issuer, refinancing an existing customer does not present additional risk, because the issuer is already responsible for advancing payments if the borrower defaults.¹⁶

B.2. Interest Rate Spreads: FHA vs Conforming Market

So far we have focused on variation in risk *within* the FHA market. Next we compare the cost of credit in the FHA market to the less-risky conforming market, estimating whether intermediation markups increased more quickly in the FHA segment, as measured either by the primary-secondary spread (Panel A of [Figure IA.28](#)) or the gain-on-sale (Panel B of [Figure IA.28](#)). We measure these objects using the same methodology as our analysis of the conforming market in Section [II](#) of the main

¹⁵As the figure shows, the fraction of loans to lower-than-680 FICO borrowers also falls but by a smaller amount. We also find similar patterns for quantities using originations from Home Mortgage Disclosure Act data—see [Figure IA.21](#).

¹⁶In fact, refinancing may even reduce risk by lowering the borrower’s monthly payment (Fuster and Willen, [2017](#)). Furthermore, a large majority of FHA refinances occur under a streamlined refinancing program that waives many requirements, including the need to conduct a property appraisal.

text, combining secondary market prices for Ginnie Mae securities from J.P. Morgan with primary market mortgage rates from the Mortgage Bankers Association.

While there is some volatility, in both cases we do indeed find a significantly larger increase in markups in the FHA market. For example, the gap in the primary-secondary spread between the FHA and conforming markets is close to zero pre-pandemic, but then increases to as high as 50-60bp. This is on top of the already large rise in the conforming spread that we document in the main text—put differently, the primary-secondary spread increased by roughly 100bp in the conforming market during the pandemic, but by up to 150bp or so in the FHA market.

The dislocation in markups between the FHA and conforming market is concentrated in 2020 and early 2021. By 2021:Q2 the gap in markups between FHA and conforming loans returned to levels similar to or even lower than pre-pandemic.

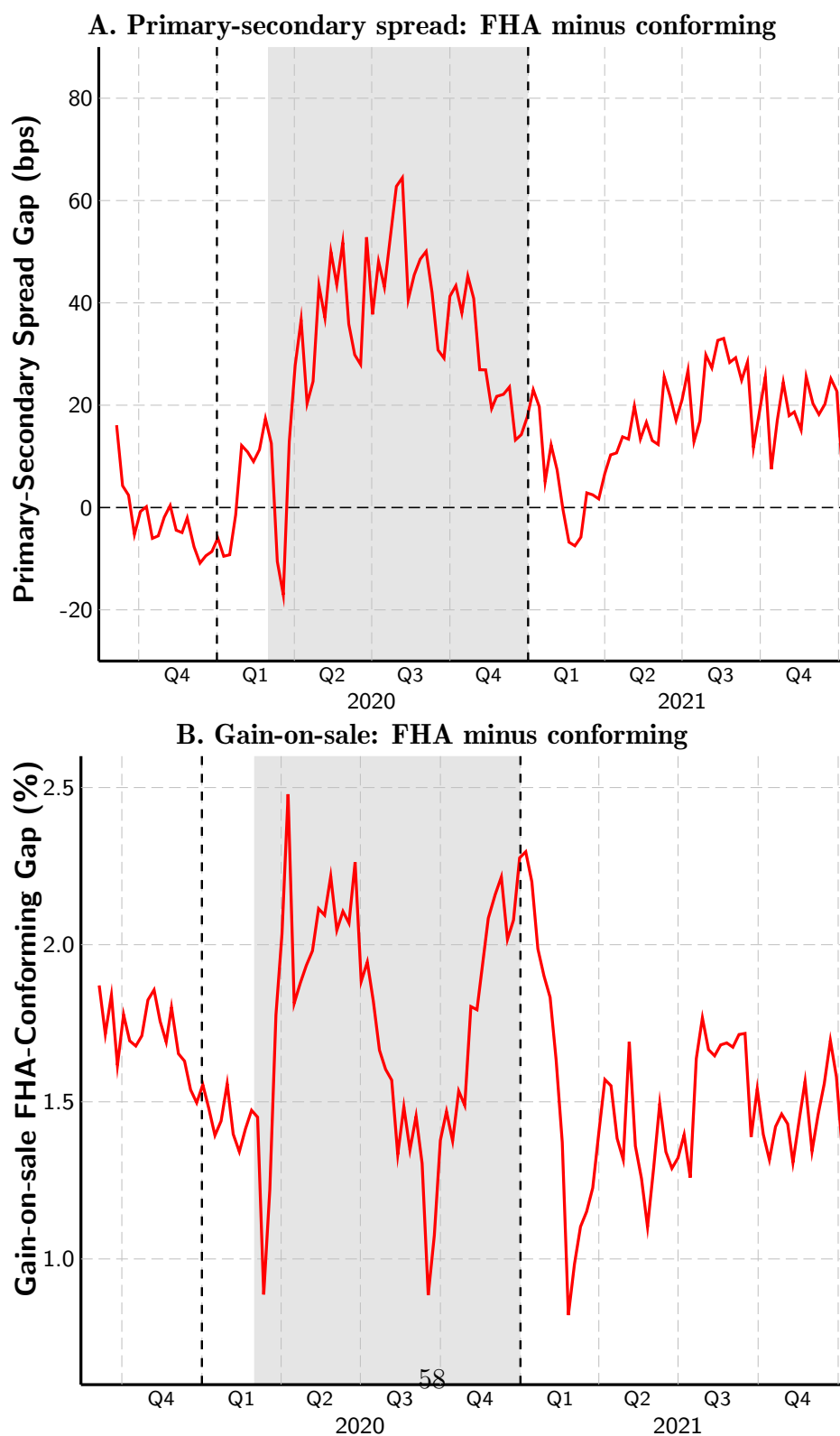
B.3. Risk vs Crowding-Out

The results seen so far are consistent with the hypothesis that the pandemic amplified the risk premium associated with FHA lending, which was then priced in the primary mortgage market. However, an alternative or complementary explanation is that FHA lending was “crowded out” due to lender capacity constraints. (See Sharpe and Sherlund, 2016 for evidence from prior periods that less creditworthy borrowers are crowded out during periods of peak demand.) This could occur if FHA loans are relatively more labor-intensive to underwrite or require more detailed interaction with the borrower.

Disentangling these two explanations is challenging because both would predict a relative decline in credit supply to riskier borrowers. However, Table IA.VI is able to make some progress by studying processing times as a proxy for the amount of effort involved in underwriting.

First, Panel A of Table IA.VI tests whether FHA mortgages are more complex to

Figure IA.28. Comparing intermediation markups in the FHA and conforming markets. Primary-secondary spread (Panel A) and gain-on-sale (Panel B) measured based on the methodologies described in Sections II.A.1 and II.C of the main text, respectively. Vertical line represents the declaration of a national state of emergency on March 13, 2020. Data sources: J.P. Morgan Markets and MBA (via Haver Analytics).



underwrite by comparing the processing time of FHA and conforming mortgages over the period from July 2019 to December 2020. We do so with and without detailed loan controls and county fixed effects, but importantly always include lender-by-month fixed effects, so we study processing time differences *within* a lender at a point in time. The results suggest that FHA mortgages indeed take a longer time to originate: prior to the pandemic, purchase mortgages took about 3-4% longer in the FHA segment than in the conforming segment, while for refinances the difference was larger, at 13-14%. The interaction term with the “pandemic” dummy indicates that the difference further increased for purchase mortgages from March 2020 onward, although the same is not true for refinances.

These patterns suggest that it is indeed possible that FHA mortgages became relatively more expensive during the pandemic because lenders wanted to allocate their scarce resources towards mortgages that were easier to process. However, Panel B of [Table IA.VI](#) finds that low-FICO conforming mortgages *also* take longer to process (compared to higher-FICO conforming mortgages), particularly during the pandemic, even though we previously showed that interest rates for such low-FICO loans did *not* differentially increase during the pandemic ([Figure 9](#) in the main text).¹⁷

This evidence from the conforming market casts doubt on the idea that lenders actively used rate spreads across loan types to shift the mix of applications towards simple-to-underwrite loans. This conclusion is also in line with the findings of Frazier and Goodstein ([2023](#)), who study HMDA data and find little evidence that capacity-

¹⁷Based on columns (1) and (3), we see that without controls other than lender-by-month fixed effects, low-FICO loans take about 2% (purchase loans) or 11% (refinances) longer to refinance. Columns (2) and (4) indicate that, with all loan-level controls, processing times for low-FICO loans increased 1-2% during the pandemic (while the uninteracted low-FICO coefficient is no longer separately identified).

Table IA.VI. Processing Times for More Complex Mortgages During the Pandemic.

Regression of mortgage processing time (measured as the difference between application date and action date) on two measures of loan complexity/risk: an FHA dummy and a dummy for credit score < 680. Estimated using loan-level HMDA data. Sample for Panel A includes FHA loans and conventional conforming loans, while sample for Panel B includes conventional conforming loans only. All specifications include lender $\times$ month fixed effects. Pandemic is defined as the period from March 2020 onwards, where the relevant time is the month of application. Loan controls in even-numbered columns include: a pre-application dummy, county dummies, debt-to-income (DTI), DTI², loan-to-value (LTV), LTV², credit score, credit score², log(appraisal amount), log(appraisal amount)², log(applicant income), log(applicant income)², applicant sex, applicant race and ethnicity, applicant age, an indicator for having no co-borrower, occupancy status, binned applicant age, and loan purpose (purchase, rate/term refinance, or cash-out refinance), where applicable. Sample period is July 2019 to December 2020. Robust standard errors in parentheses clustered at the lender level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

A. FHA Mortgages (vs. Conventional Conforming)

	log(processing time)				processing time (days, winsorized)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FHA	0.041*** (0.007)	0.029*** (0.004)	0.128*** (0.039)	0.144*** (0.031)	0.621 (0.440)	0.703*** (0.227)	5.636*** (1.349)	5.959*** (1.134)
FHA $\times$ pandemic	0.015*** (0.003)	0.018*** (0.004)	-0.049 (0.044)	-0.023 (0.032)	1.272*** (0.215)	1.368*** (0.226)	-0.905 (2.357)	0.308 (1.770)
N	5315040	5313561	9568310	9566126	5315040	5313561	9568310	9566126
Mean Y	3.73	3.73	3.90	3.90	49.95	49.95	56.83	56.83
Origination type	Purchase	Purchase	Refi	Refi	Purchase	Purchase	Refi	Refi
Loan Controls	N	Y	N	Y	N	Y	N	Y
County FE	N	Y	N	Y	N	Y	N	Y
Lender $\times$ month FE	Y	Y	Y	Y	Y	Y	Y	Y

B. Low-Credit-Score Mortgages in the Conventional Conforming Segment

	log(processing time)				processing time (days, winsorized)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
FICO < 680	0.021*** (0.005)		0.111*** (0.007)		0.599* (0.326)		5.924*** (0.231)	
FICO < 680 $\times$ pandemic	0.001 (0.004)	0.013*** (0.003)	0.004 (0.003)	0.019*** (0.003)	-0.036 (0.225)	0.808*** (0.191)	1.589*** (0.272)	2.123*** (0.229)
N	4181063	4179846	8828705	8826559	4181063	4179846	8828705	8826559
Mean Y	3.73	3.73	3.90	3.90	50.10	50.09	56.93	56.93
Origination type	Purchase	Purchase	Refi	Refi	Purchase	Purchase	Refi	Refi
Loan Controls	N	Y	N	Y	N	Y	N	Y
County FE	N	Y	N	Y	N	Y	N	Y
Lender $\times$ month FE	Y	Y	Y	Y	Y	Y	Y	Y

constrained lenders control demand by raising prices on loans to marginal borrowers.¹⁸

B.4. FHA Analysis: Summary

We find a more pronounced contraction in credit supply in the FHA market than the conforming market during the pandemic, reflected in a larger spike in mortgage interest rate spreads and gain-on-sale, as well as a decline in the volume of lending to the riskiest FHA borrowers and the number of lenders serving that segment.

While our results suggest that FHA credit supply contracted because of the higher risk in that segment, an alternative possibility is that lenders increased FHA markups because they instead wanted to focus on easier-to-originate conforming loans. This explanation is not fully consistent with patterns in the conforming segment though—high-risk conforming loans also seem more complex-to-underwrite but did not carry an interest rate premium during the pandemic. We view this as suggestive, although not decisive, evidence that high FHA markups in 2020-21 are at least partially due to risk.

In any case, under either interpretation, a key takeaway from our findings is that “government guarantees are not enough.” We observe significant differences in credit supply between FHA loans and conventional conforming loans during the pandemic episode, even though both types of loans carry a government-backed credit guarantee.

¹⁸Instead, Frazier and Goodstein (2023) present evidence suggesting that when capacity constraints are binding, lenders may ration supply to marginal borrowers before the application stage, perhaps by “not returning the calls” of such borrowers.

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